3D-Printed Medical Robotic Exoskeleton Glove for Home Rehabilitation

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Abstract

Partial paralysis of the upper limbs is an increasingly prevalent issue among elderly individuals in Hong Kong, especially those who have experienced strokes, spinal cord injuries, or muscular dystrophy. As the proportion of elderly persons is projected to increase in the coming years, this rapidly aging population places immense pressure on healthcare infrastructures involved in the management of rehabilitation services for age-related conditions such as stroke. Research shows that physical therapy can restore upper limb function with consistent repetition of motor impairment rehabilitation exercises. However, current hand rehabilitation devices are time-consuming, expensive, and difficult to access due to travel and time constraints. A wire-driven 3D printed robotic hand is thus proposed due to the properties of being customizable, portable, and allowing control of individual finger movements during rehabilitation exercises or daily tasks.

This study presents an improved design for home medical hand rehabilitation devices that can independently control the displacement of each phalanx to perform hand rehabilitation exercises or daily tasks. The modular design of these 3D printed flexible thermoplastic polyurethane (TPU) exoskeleton gloves makes it possible for its customization, resulting in increased comfort for individuals with a heterogeneity of hand sizes and are also lightweight and portable outside of clinical settings. These gloves allow for a two-way motion of all 5 fingers by transforming the rotatory motion from the servo to a linear motion that moves the fingers on a horizontal plane through the pulling and releasing of strings. This device allows for both an automatic object detect-and-grab system, allowing the glove to automatically grab onto everyday objects when they're within a certain distance, and a mirroring system that imitates the movement of the patient's other hand to their injured hand wearing the exoskeleton glove.

Keywords:

Partial paralysis, stroke, muscular dystrophy, motor impairment, robotic exoskeleton glove, rehabilitation exercises, 3D printed

I. Introduction

Stroke in Hong Kong

Outside of traditional work-caused hand injuries, another cause contributes significantly to the occurrence of upper limb motor impairment: Stroke. Stroke is the fifth leading cause of death in Hong Kong, with 3057 registered stroke-related deaths solely in the year 2022. Hong Kong University researchers also indicated in a study that there's currently an upwards trend in the number of "crude incidents of young stroke", with the number of incidents increasing from 39.1 to 48.3 and to 55.7 per 100,000 people in years 2001, 2011 and 2021 respectively [11].

Young individuals in the above study were defined as 18 years old to 55 years old. However, stroke can also significantly affect older populations. Research published in the National Library of Medicine indicates that risks of stroke doubles every decade after the age of 55 [25], which can be attributed to the increased prevalence of factors such as hypertension, diabetes, and cardiovascular diseases in older populations. Coincidentally, the Census and Statistics Department of Hong Kong indicated in a press release that "the proportion of elderly persons aged 65 and over is projected to increase from 20.5% in 2021 to 36.0% in 2046", with the trend primarily attributed to the rising life expectancy and decreasing number of births in Hong Kong [12]. However, this rapidly aging population can have unexpected detrimental effects on the healthcare system, in particular, placing immense pressure on healthcare infrastructure involved in the management of rehabilitation services for age-related conditions such as stroke. Additionally, the increased number of incidents of stroke will also increase the demand for long-term rehabilitation programs and devices. Since post-stroke recovery often requires intensive rehabilitation programs which can last for months or even years, this further exacerbates the demand for innovative solutions in the rehabilitation sector.

This paper presents the design of a portable and customizable hand rehabilitation exoskeleton glove. Studies have shown that hand paralysis is especially common in people who have experienced a stroke, with around 80% of individuals who have suffered a stroke having experienced some degree of upper limb paresis [14]. The degree of motor impairments may vary, ranging from diminished sensory perception to loss of dexterity and, in severe cases, complete paralysis. Hand injuries also account for ⅓ of all injuries at work, ⅕ of permanent disability [17], and can significantly reduce a patient's quality of life. Hence, optimizing the recovery of upper limb functions is vital for regaining independence in daily activities, especially for essential self-care activities like dressing, feeding, and grooming [16].

Recent developments in robotic hand rehabilitation devices have great potential to address these issues. Due to the devices' ability to independently perform rehabilitation tasks in a home setting without a therapist, it facilitates more frequent and consistent therapeutic engagement in a relatively more comfortable environment outside of clinical settings. Subsequently, these devices have the potential to expedite motor function recovery through heightened engagement with tasks that mirror the activities of daily living. This helps avoid both the motivation struggles associated with exercise and the financial burden of therapist-led rehabilitation, allowing the patient to undergo rehabilitation more frequently and regularly. Another benefit of these devices is the ability to perform therapeutic exercises with precision, ensuring the proper tissues are being exercised and consequently reducing the risk of injuries during the recovery process. These rehabilitation devices can be broadly categorized into two types: end-effector and exoskeleton.

End-effector Rehabilitation Devices

Unlike exoskeleton devices, end-effector rehabilitation devices have actuators over the distal interphalangeal (DIP) joints, rather than a glove over the entire finger. The EFRR (end-effector finger rehabilitation robot) is an example of such a device. The device (Figure 1.1) has several series of elastic actuators located beneath the fingers of the hand, mimicking the movement of finger joints through a fixed pulley-track module that controls the flexing and extending of the patient's finger [21]. While the device achieves the ability to flex and extend fingers precisely to perform rehabilitation movements, there are several limitations. By employing complex actuators and sophisticated hardware, the device is particularly large and costly, hence limiting the potential for remote rehabilitation that has been shown to enhance motor recovery. Additionally, to protect its delicate instruments, the EFRR uses a rigid frame. When combined with its complex and highly sophisticated interior mechanisms, the overall mass of the EFRR can pose significant challenges. Such issues include the high costs of transportation and manufacture, which reduces the device's portability and accessibility to patients with lower economic capabilities.

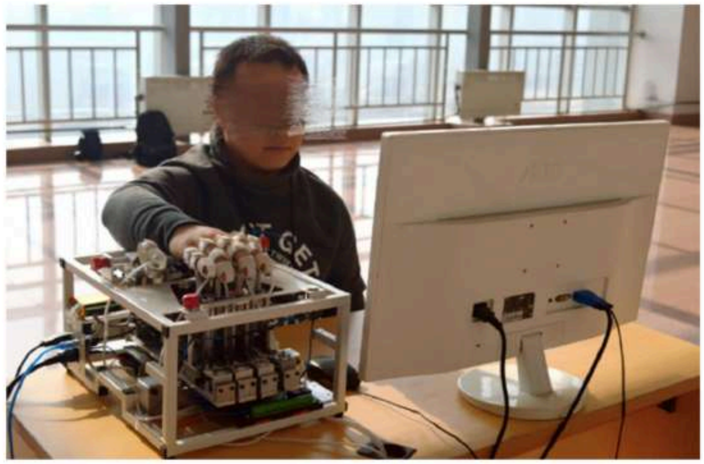

Figure 1.1: Prototype of the EFRR: end-effector finger rehabilitation robot [21].

Another hand rehabilitation device that utilizes the end-effector model is Amadeo by Tyromotion [23]. After the hand and forearm are placed on a platform connected to the main unit, the wrist is secured with Velcro to prevent movement of the elbow and shoulder. Subsequently, every finger is strapped separately on magnetized rails sliders, enabling finger flexion and extension. The device also features a screen facing the patient which allows for the assessment of spasticity through games, tasks, or specialized programs in an interactive format, thereby providing supplementary benefits for the recovery of the patient’s fine motor skills through heightened engagement with the rehabilitation tasks.

However, there are several notable drawbacks to the Amadeo. Primarily, the consequence of such a highly technical and large device (Figure 1.2) is the necessity of a physical therapist’s presence during its use, thus constraining its capacity for independent utilization by patients. Additionally, due to the device's complex hardware and large dimensions, the Amadeo is marketed at $35,611 USD, rendering the device financially inaccessible for patients with a lower socioeconomic status.

Figure 1.2: Robotic Therapy with AMADEO [3].

Exoskeleton-Based Rehabilitation Devices

Hand exoskeleton-based rehabilitation devices have been extensively studied due to their ability to provide rehabilitation for a patient's fine motor functions in their hand by encompassing each individual finger, a feature that is particularly beneficial for patients with advanced injuries [15]. These types of devices have become increasingly popular for commercial production as well as scientific research.

An example of such devices is the SaeboGlove by Saebo. This device utilizes elastic bands and contains customisable tensioners to assist in finger extension and flexion. Primarily made of soft fabrics, this exoskeleton based glove is soft to the touch and doesn't offer any sharp or rigid parts that may damage the human skin. Additionally, it offers a variety of glove and tensioner sizes, allowing patients with a heterogeneity of hand measurements to comfortably utilize the glove for rehabilitation [19]. However, to return a patient to full motor functions, all three types of rehabilitation, active, active assisted and passive are required [15]. The SaeboGlove, in contrast, is limited to only one mode of recovery: active rehabilitation, as shown with the lack of actuators. By only offering active rehabilitation, it restricts the device's ability to comprehensively address the wide spectrum of patient's needs with varying levels of impairment.

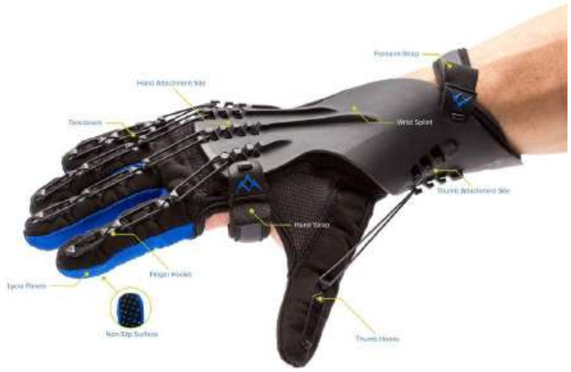

Figure 1.3: The SaeboGlove [18].

Another exoskeleton-based hand rehabilitation device is the CyberGrasp [7]. This lightweight device uses 5 actuators to pull strings that are attached to each finger, allowing flexion and extension through a linkage system. This device also utilizes a VR-based therapy that simulates grasping a computer-generated 3D object, engaging with the user while also providing haptic feedback. A downside of incorporating VR, actuators, and a multi-finger linkage system means this type of device would likely be more expensive than simpler options, limiting widespread adoption and access. Additionally, the reliance of a VR system requires a separate VR system, software and other equipment and could negatively impact its portability.

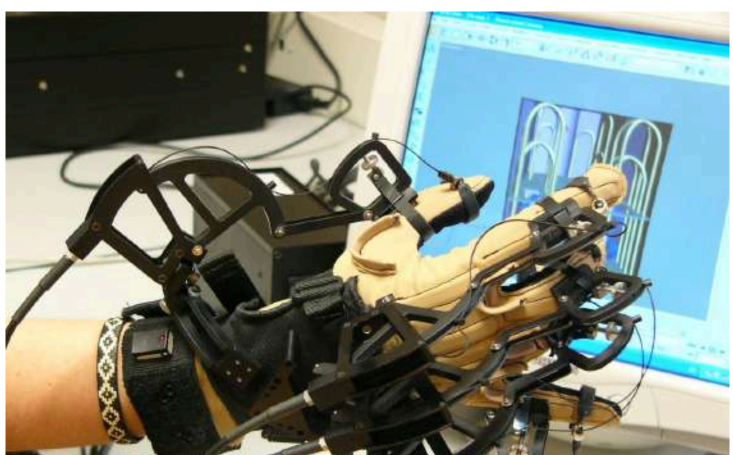

Figure 1.4: CyberGrasp – an example of an exoskeleton haptic device [7].

While existing hand rehabilitation devices have shown promise, limitations remain in cost, customizability, and the ability to use the device without a physical therapist present.

Commercial systems employing complex actuators and sensing can be bulky, uncomfortable to wear for extended periods, and costly due their sophisticated hardware. This limits the potential for intensive, home-based therapy that has been shown to enhance motor recovery. Additionally, many existing exoskeletal and end-effector devices utilize a rigid structural frame, an inflexibility design that limits the accessibility of such systems across user populations with diverse anthropometric traits.

This study developed a polymer-based, wire-drive, light wearable exoskeleton glove. The glove is 3D printed, utilizes a flexible thermoplastic polyurethane material known as TPU, with its modular and customizable design allowing the glove to be designed for each individual user's unique hand size and dimensions through an automated process, improving comfort. TPU has shown to be both flexible, 3D printable and possesses the necessary physical properties such as having a coefficient of static friction that exceeds 1.00 [20], preventing slipping when securely grasping objects. Additionally, each phalanx can be independently controlled through a system of bend sensors on the uninjured hand, the bi-directional motion, the flexion and extension of each digit corresponding to the servos that pull string on each finger on the injured arm, mirroring the motion and assisting in a range of motion exercises. The incorporation of an automatic object detection-and-grab system enhances the patient's engagement with the rehabilitation exercises, allowing the patient to both undergo rehabilitation while independently performing daily activities. Lastly, due to the 3D printed rehabilitation glove's soft, comfortable and lightweight material and design, the glove is extremely portable and can be used outside the clinical settings.

II. System Overview

1. Principle Analysis

A. Design of the glove

The rehabilitation glove comprises several 3D designed modules: The finger, thumb, and the main palm. These modules conjointly create a system that funnels 10 individual wires, one for each finger, from the finger exoskeletons to the servo motors that actuates the flexion and extension motion of each finger. The servo motors are located in another 3D designed casing strapped onto the bicep that protects the motors and allows for a structured system that guides the wires through small plastic tubes along the arm and into the hand exoskeleton design. A smaller capsule is attached onto the motor casing, protecting the electrical components used to control the motors.

The lengths and degrees of freedom (DoFs) of a hand exoskeleton rehabilitation device should match the anatomical structure of the finger joints. The joints of a human finger, excluding the thumb, include the metacarpophalangeal (MCP) joint, the proximal interphalangeal (PIP) joint, and the distal interphalangeal (DIP) joint. As not all joints need to be actuated for basic hand rehabilitation exercises, an exoskeleton design for finger rehabilitation could consider only actualting only the MCP and PIP joints, which are responsible for the flexion and extension movements, and are critical for grasping objects. This reduces the cost and complexity of the device while also maintaining the full range of motion in the recovery of key motor functions, the flexion, and extension of each digit, which is essential in the daily activities of patients. The ability to achieve thumb-finger opposition is also a crucial element of object manipulations [2] of humans and is imperative to regaining independence in daily tasks for patients. As the function and anatomy of the thumb is quite different from the other 4 fingers in the same hand, containing the Trapeziometacarpal (TM), the metacarpophalangeal joint (MCP) and the interphalangeal joint (IP), a specialized thumb exoskeleton design is required to fulfill its function.

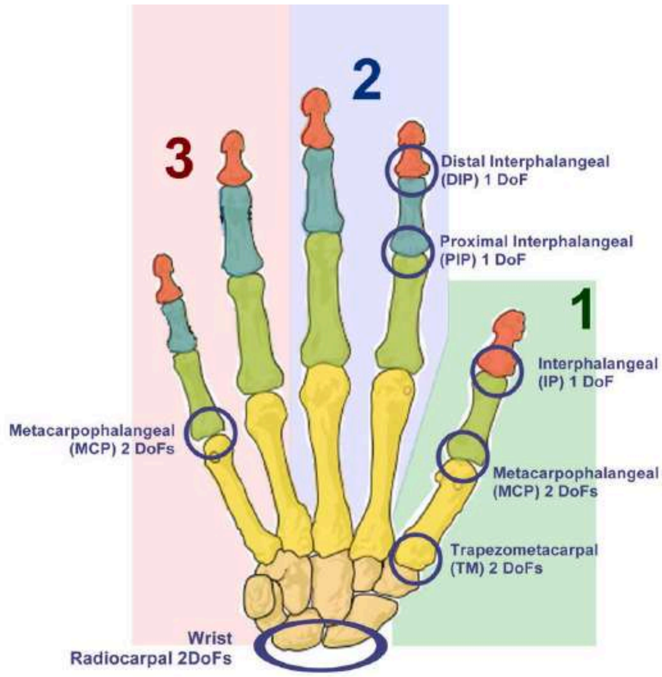

Figure 2.1: Human hand structure. The joints in the human hand and the relative DoFs. The carpometacarpal joint in the little finger is omitted [13].

Due to the unique anatomy of the human hand, a one-size-fits-all hand rehabilitation exoskeleton design would likely not optimally fit or functionally support the natural movements of all users. To address this issue, a 3D printed design was proposed to have more control over the specifications of the exoskeleton that's customized according to the user's specific hand scan data. Additionally, the usage of thermoplastic polyurethane (TPU) allows for a more flexible and comfortable design, reducing the discomfort compared to a rigid design. The TPU also provides a lighter weight design compared to more rigid alternatives, making the device more portable and convenient for patients to use both within clinical settings and in their daily activities outside of therapy sessions.

A wire-driven system was chosen to reduce the weight and prioritize the ease of movement of the hand. The system converts the rotational motion of servo motors to the tension of the wire, causing its linear push and pull motion and allowing the flexion and extension of each individual digit of the hand. Additionally, as the flexion and extension of the finger can be controlled by the clockwise and anticlockwise turning of the motor that pulls and releases the wire, only 1 motor per finger is required, reducing the number of actuators and therefore the weight and bulkiness of the glove. However, to perform the full range of motion required for hand rehabilitation for finger flexion and extension in daily activities, actuators need to provide sufficient force.

B. Theoretical calculations

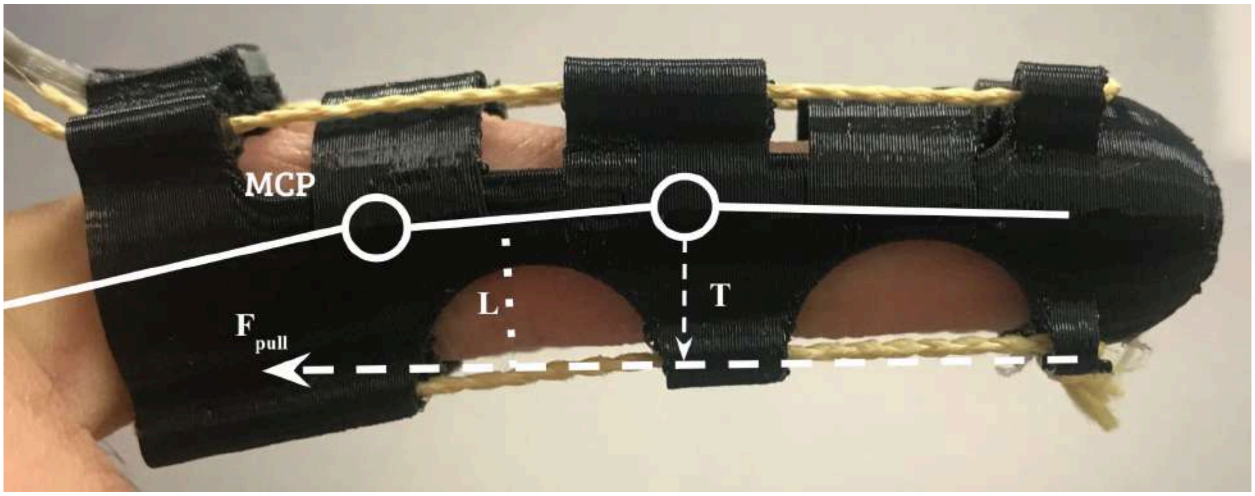

Figure 2.2: Finger exoskeleton prototype on ring finger.

The pulling force required for each finger's motor can be calculated by the following:

$$\tau = L \times F_{pull}$$

Where F_{pull} is the linear rotation force provided by the servo motor, given as 12kg/cm when connected to 11.1 volts (parameter shown on product website). The distance L is the distance between the joint and the string on the finger exoskeleton. For example, L is measured to be 8.5mm in the ring finger shown in figure 2.2. and τ represents the torque of the finger is therefore calculated to be $(12\times 10^{2})N \times 8.5 \times 10^{-3} m \approx 10\ N \cdot m$

2. Structural Design

The design of the exoskeleton glove is split into 3 main sections: fingers, palm and motor box. The wire routing system runs from the motor box, through 5 plastic tubes, the palm and into the finger exoskeletons. By isolating the heavy motor box to the bicep, weight is distributed away from the fingers to maximize dexterity and comfort.

A. Fingers exoskeleton

Basic finger model

A human finger can roughly resemble a cylindrical cone-like shape when seen from the side. Specifically, the phalanges gradually narrow from the metacarpophalangeal (MCP) to the distal interphalangeal (DIP) joint, forming a conical frustum with the distal phalanx and a semi-spherical portion that encloses the fingernail. Of the three joints, the MCP joint is typically the largest when seen from the back of the hand. However, due to the difficulty in accurately capturing the width of the MCP joint through a camera scan, the width of the frustum's base, which plays a large role in determining the size of the exoskeleton, should be proportional to the width of the PIP joint rather than the MCP joint. An extra 2 mm is added to the exoskeleton base width, accounting for a margin of error while 3D printing and leaving a gap between the finger and the exoskeleton to improve comfort. The height of the model is equal to the distance from the tip of the finger to the top of the MCP joint. To serve as a barrier for the skin while also ensuring flexibility, a width of 1.5 mm was chosen for the exoskeleton design. The thumb design is completed similarly with the MCP and IP joints, with the exception of one less joint opening due to the thumb's unique anatomy.

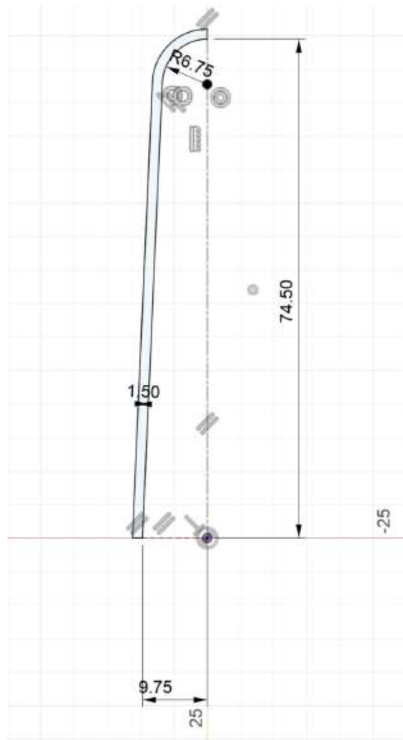

Figure 3.1: sketch on Fusion 360 of basic finger model.

Wire tube

To guide the wire towards the tip of the finger, a cylindrical tube like structure with a thickness of 2mm is constructed on a plane perpendicular to the natural slant of the exoskeleton, extending from its base to a 5 millimeters away from its tip. As the tube needs to be securely integrated into the basic finger model, part of the tube and the model overlap, a fillet forming at the edges between these two structures is created to maximize the structural integrity. The tubes are subsequently structured at a 54 degree angle and mirrored vertically across the center of the circle, ensuring that the tug of the string maintains a balance and doesn't lean towards one side or another during flexion. The two tubes are mirrored again horizontally to help facilitate the extension of each digit (figure 3.3). As the two tubes on the dorsal surface perform the same motion, a single wire can be passed through them and tied to the servo motor, thus reducing the amount of wires and motors required. The same applies to the two tubes on the palmar surface, as they can also be tied to the same motor on the opposite side to perform the opposite motion(figure 3.2). During flexion, the motor pulls the palmar wire while releasing the dorsal wire, bending the finger. For extension, the action is reversed - the dorsal wire is pulled and the palmar wire released. This synchronized pulling of one wire surface and releasing of the other dramatically reduces the weight and size of the motor box that's attached to the biceps due to a reduced number of motors and servos required.

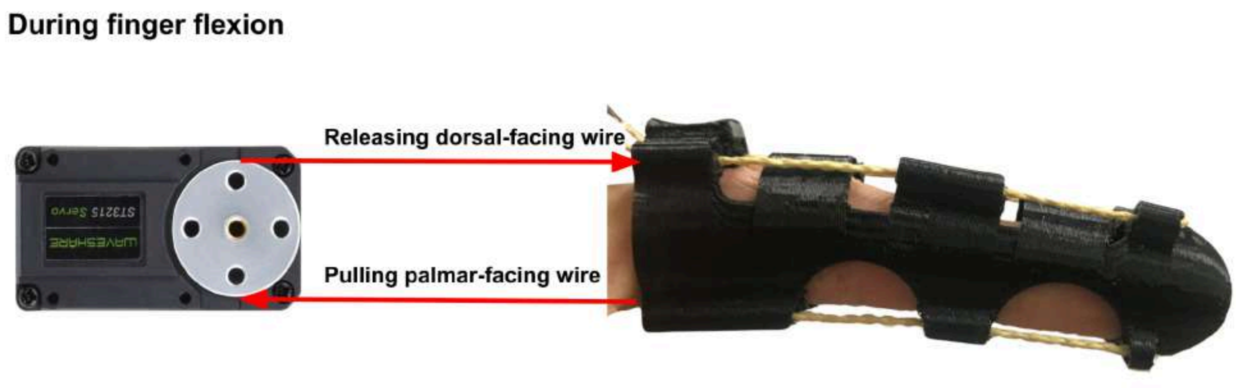

Figure 3.2: The motion of wires during finger flexion.

Figure 3.3: The basic finger model designed in Fusion 360.

Joint cutouts

To further enhance comfort and mobility, interior semi-circle shaped cutouts, referred to as joint cutouts, matching the contour of each joint are incorporated below the PIP and DIP joints, preventing inward folding of exoskeleton material during flexion that could push against and cause impingement for the delicate skin below the joint. A similar larger rectangular shaped cutout is integrated above the PIP and DIP joints, allowing a wider range of unrestricted movement of the finger (figure 3.4).

Guard flap

To further ensure an equal distribution of force, guard flaps are added to serve as a barrier that keeps the string from rubbing against the skin of the finger during flexion and extension of the fingers in the exposed joint cutouts. These take the form of rounded prismatic flaps, and are positioned on both sides of the joint cutouts located on the dorsal surface of the hand. During finger motion, the guard flaps serve to direct and isolate the moving wires, preventing direct contact and friction with delicate skin near the joints.

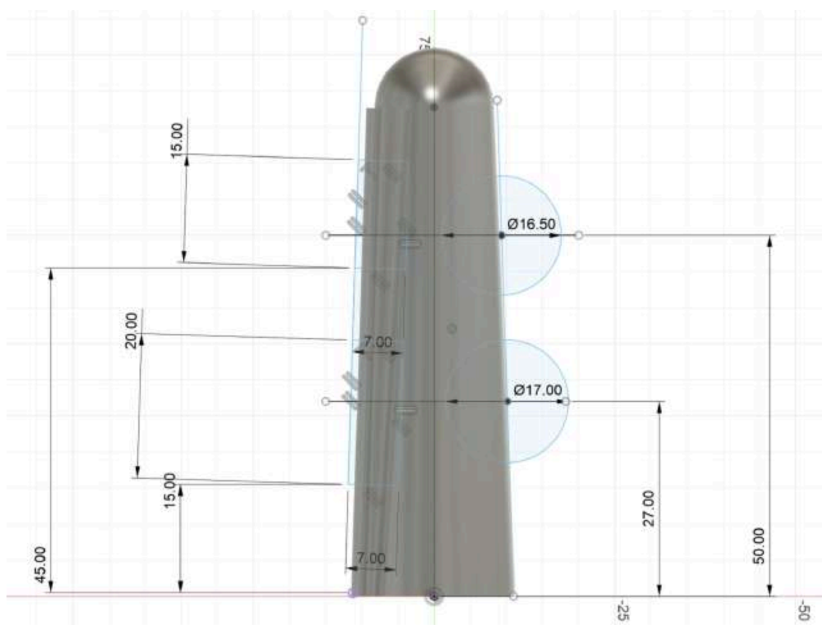

Figure 3.4: Measurement of the position of guard flaps and joint cutouts according to the researcher's own finger anatomy.

B. Guard flaps

The guard flaps (figure 3.5) serve as a flexible transition between the finger model and the palm. Attached through plugging it into a rectangular hole at the base of the finger exoskeleton, these 1.5 mm rectangular pieces contain an array of triangular patterns to maximize their elasticity without compromising structural integrity. The triangular patterns take the form of equilateral triangles with sides of 5 mm each, arranged in staggered columns along the length of each flap. During flexion, the flaps transition from being in a bent position to lying flat against the skin of the MCP joint.

Figure 3.5: Sketch of guard flap located above the knuckles of the hand.

C. *Palm exoskeleton*

The purpose of the palm exoskeleton is to protect the skin while also creating a fixed pathway for the wires to travel from the motor box to the finger exoskeletons. The basic palm model is created through measuring the two ellipses of the wrist and the area around the MCP joints and lofting them together, creating a smooth conical shape. The guiding tubes for the dorsal wires, with the thickness of 2mm to smoothly redirect wires, are created in a similar manner, with the interior radius being 1.5 mm to provide adequate space for the movement of the wires. To securely strap the palm exoskeleton onto the hand, a vertical cut was made that allows the exoskeleton to wrap around the hand comfortably. Two Velcro adjustable straps were also incorporated, one connected through the thenar eminence and the other tying the vertical slit together. To allow natural thumb movement, an elliptical cutout from the Trapeziometacarpal (TM) joint to the tip of the thumb was made. The palmar wires are also funneled through an oblong-like shape located below the palm that allows the tube connected to the motor box to merge with the palm exoskeleton. To facilitate the wire's movement, a second tube with the length of 20 mm is connected. Five connection hollow rectangular ports were also integrated into the palm exoskeleton to facilitate the transition to the finger exoskeletons. To reduce the amount of supporting material created during 3D printing which can be a hassle to remove afterwards, the base of the ports were altered to be angled at 45 degrees, achieving the necessary stability to be printed without support. Due to the position and movement of the thumb being different than the other four fingers, two separate funnel for the thumb is created, one at the trapeziometacarpal (TM) joint responsible for extension, and the other between the MCP joint of the pinky and the wrist to created the thumb-opposition movement that's essential for object manipulation in daily tasks.

Figure 3.6: The palm exoskeleton design modeled in Fusion 360.

D. Motor Box

Printed with PLA to provide a strong structural support and protection for the 5 motors in the interior, one motor for each finger, the motor box is strapped securely on the bicep of the user. The box consists of 2 layers of an exterior wall and a motor board stacked upon each other to reduce its length. The motor board comprises a staggered 3-socket configuration with sockets located a distance of 15 mm away from each other, each with four small screw holes to attach it securely and a semi-circular cutout allowing the motors to rotate freely. To strap the board to the bicep, four ports are created to strap two Velcro bands onto the user, with the now secured motor board connected to the exterior walls through 4 screws. The second component of the motor box is the thick exterior walls, which has 6 front-facing circular holes to accommodate the pull and release of strings in the front and back of the hand model that comes in pairs. As there are 3 sockets that can attach 3 gears and each gear can pull and release 2 strings, there are 6 [2x3] holes a radius of 2mm cutout from the walls. A square cutout for the passage of wires is also included in the second motor board to power and transfer signal to the servo motors internally in the box.

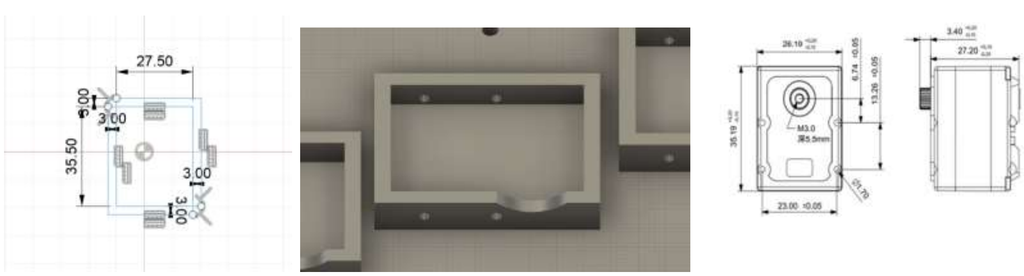

Figure 3.7: Bus Servo socket dimensions [10].

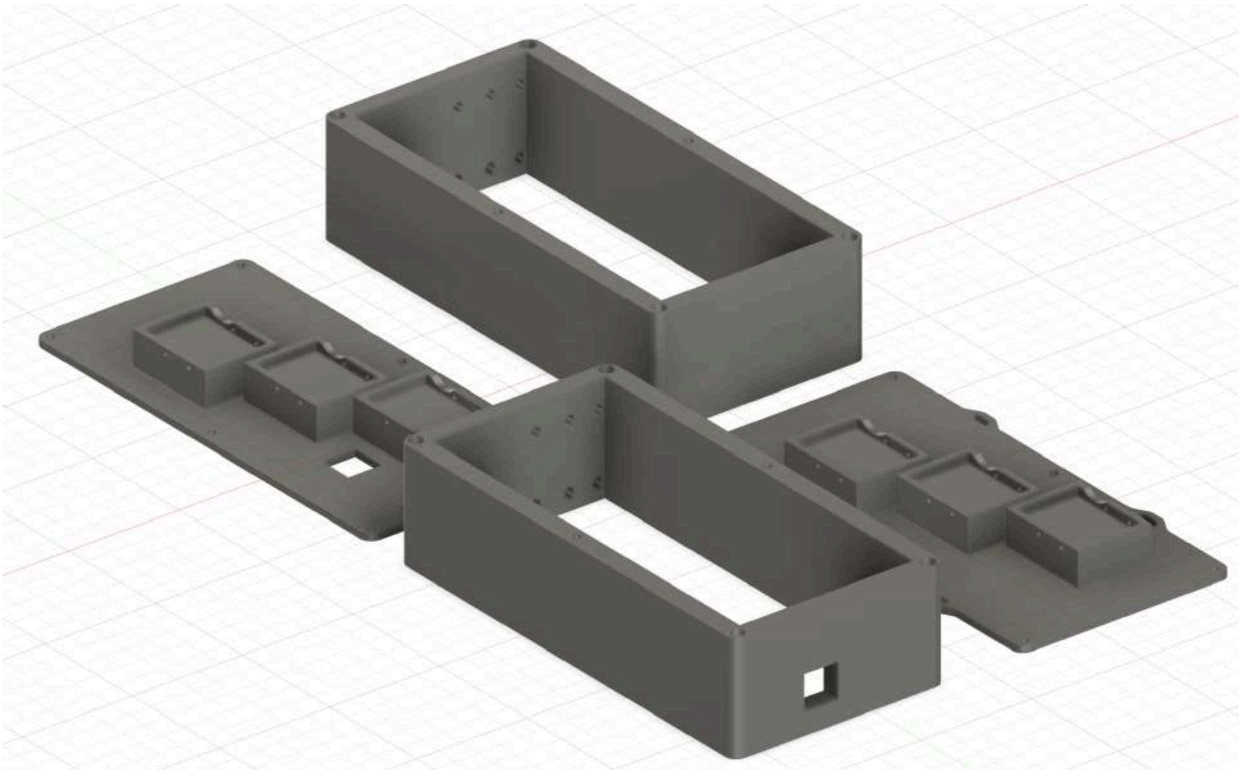

Figure 3.8: Motor box model.

E. Joint length adjuster

The joint length adjuster is a small “X”-shaped 3D printed part that reduces the length of string between the finger exoskeleton and palm exoskeleton through tangling. Positioned under the finger exoskeleton, this part provides a convenience for the user to shorten string length and thus maintain string tension, allowing the proper pulling and releasing of the palmar and dorsal facing wires respectively, enabling the flexion and extension of the finger exoskeletons and effective rehabilitation.

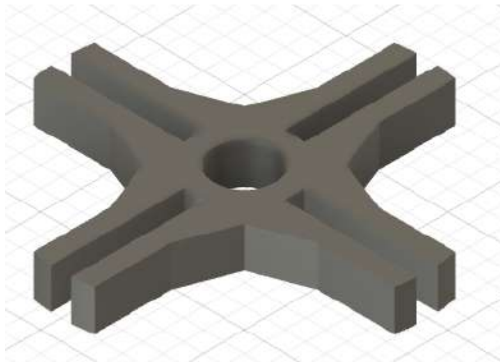

Figure 3.8: Joint length adjuster design.

F. Attachment Bracket for K210 Visual Module and Ultrasonic Sensor

To integrate the camera module into the hand exoskeleton, an attachment bracket was used. For the K210 to accurately detect the objects in daily life, it requires an unobstructed view of the area in front of the thenar space. As the wrist possesses the quality of dynamically changing its position based on the hand's rotation, it was chosen as an anchor point for the attachment bracket. To account for the change in angle between the wrist and the area above the thenar space, the camera attachment bracket was positioned at a 35° declination. Due to the complexity in the camera positioning, the degree of rotation around both the x and z axis was determined manually and secured using a bolt-locking mechanism, allowing the camera to be fine tuned to maximize the accuracy of object detection. Additionally, an additional in front of the camera bracket was created to attach the ultrasonic sensor, allowing for a clear line of sight for precise distance detection.

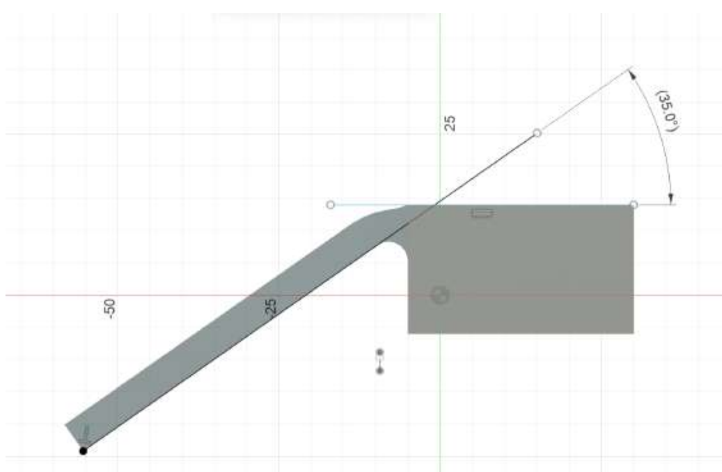

Figure 3.9: attachment bracket angled at 35° to account for the angled position of the thenar space relative to the wrist.

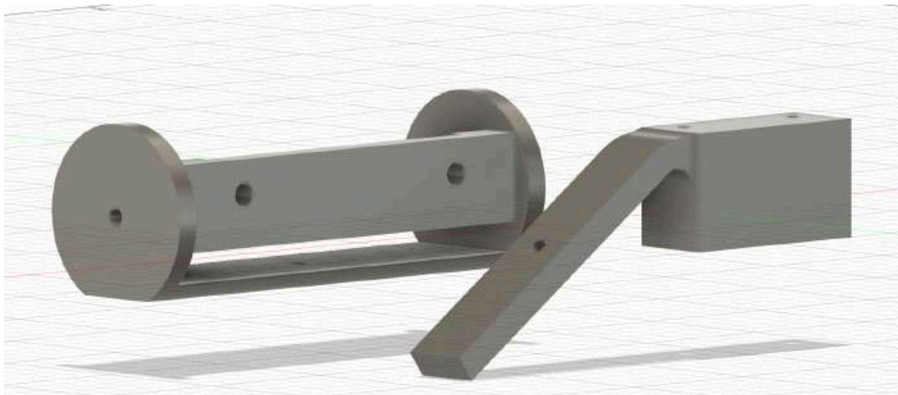

Figure 3.10: Joint length adjuster modeled in Fusion 360.

3. Electronic Design

Figure 4.11 shows a diagram of the circuit connection diagram of the rehabilitation glove and the mirroring glove. The angle of the mirroring hand is measured through flex sensors, linked to an Arduino Nano board, which connects to a Serial Bus Servo debugging board, powering all 5 servo motors. Additionally, the image recognition signal from the K210 module and the distance detected from the ultrasonic sensor is able to be sent to the arduino board, allowing for the timely actuation of servos.

A. Arduino Nano

An Arduino Nano board development board shown in Figure 4.1 is used as the main control board for this project, containing 8 analog input pins which are adequate to read data from the 5 flex sensors attached to the wired glove [4]. The Arduino Nano board was chosen as it possesses the necessary quantity of input and output pins while also being lightweight and significantly smaller than other microcontroller boards, reducing the overall dimensions of the design and increasing portability, and is thus appropriate for the design of a wearable rehabilitation exoskeleton glove. Additionally, the Arduino Nano is also a microcontroller board that is breadboard friendly and open source, and is therefore more convenient to use and cheaper than other boards.

Figure 4.1: Arduino Nano board.

B. Arduino Flex Sensor

The Arduino Flex Sensor used in this project, shown in Figure 4.2, is a common flex sensor that acts like a variable resistor, increasing and decreasing its resistance based on the degree that it bends. The flex sensor's position can range from being straight to being bent at over 90°, its resistance similarly shifting from approximately 10 kΩ to 22 kΩ respectively [8].

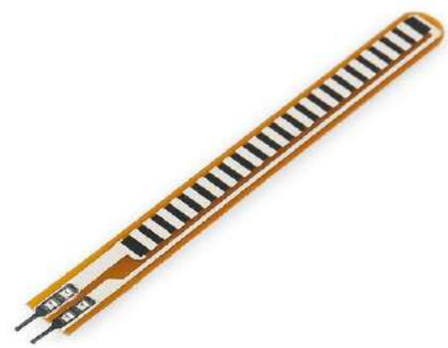

Figure 4.2: Arduino Flex Sensor [9].

C. HX-12H Bus Servo

Figure 4.3 shows the HX-12H Bus Servo used in the design. This servo motor was chosen for its ports that allow for a series connection between other servos, a clear and concise way to link several motors together. Additionally, the bus servo also allows the program to control its degree of rotation within a 240° range, achieving rehabilitation functionalities of the flexion and extension of the finger exoskeleton through the pulling and release of dorsal and palmar- facing wires.

Figure 4.3: Bus Servo gears [10].

D. Resistor

5 metal film resistors (Figure 4.7), one for each finger, was used to create voltage divider circuits shown below:

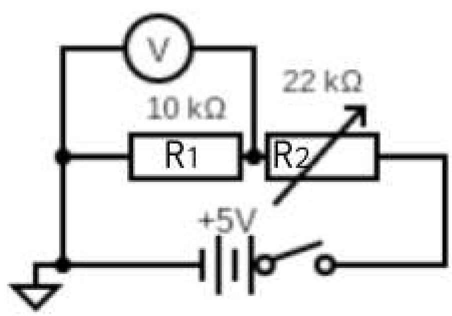

Figure 4.4: Voltage Divider Circuit.

The current of the circuit can be given as:

$$I = \frac{U}{R_1 + R_2} \qquad \text{…………(1)}$$

Where:

U = Potential difference of circuit

R_1= Fixed resistor

R_2= Variable resistance of the bend sensor

The voltage measured by the voltmeter for R_1 can also be expressed as:

$$V_1 = I \cdot R_1 \quad \text{...........(2)}$$

Where:

V_1 = Voltage measured by the analog pin of the Arduino Nano

R_1= Fixed resistor

I= Current of circuit

Thus, by substituting equation (1) into (2), the voltage measured by the arduino analog pin can be denoted as:

$$V_1 = \frac{U \cdot R_1}{R_1 + R_2}$$

The purpose of this voltage divider circuit is to measure the change in the resistance of the flex sensor, which can be enhanced through investigating the optimal fixed resistance of R_1 that yields a circuit that maximizes the change in potential difference of V_1, measured by the voltmeter, between the maximum and minimum resistance of the flex sensor, which are 10kΩ and 22kΩ respectively. As U, the operating voltage of the circuit is fixed at 5 volts, the operating voltage of an arduino nano board, and the maximum and minimum of the variable resistor R_2 is known, the relationship between R_1 and the flex sensor's sensitivity can be expressed as:

$$y = \frac{5x}{x+10000} \quad \text{...........(equation 3, Figure 4.5 in purple)}$$

$$y = \frac{5x}{x+22000} \quad \text{...........(equation 4, Figure 4.5 in red)}$$

Where equation 3 represents the relationship when the flex sensor is straight and equation 4 demonstrates the relationship when the flex sensor is bent. Thus, the difference between these two graphs is the rate of change of the voltmeter's sensitivity to shifts in the resistance of the flex sensor, which can be described as the following:

$$y = \frac{5x}{x+10000} - \frac{5x}{x+22000} \quad \text{...........(equation 5, Figure 4.5 in blue)}$$

By graphing the above equation into Desmos, the following bell curve is shown (Figure 4.5), its vertex being the point (14832, 0.973):

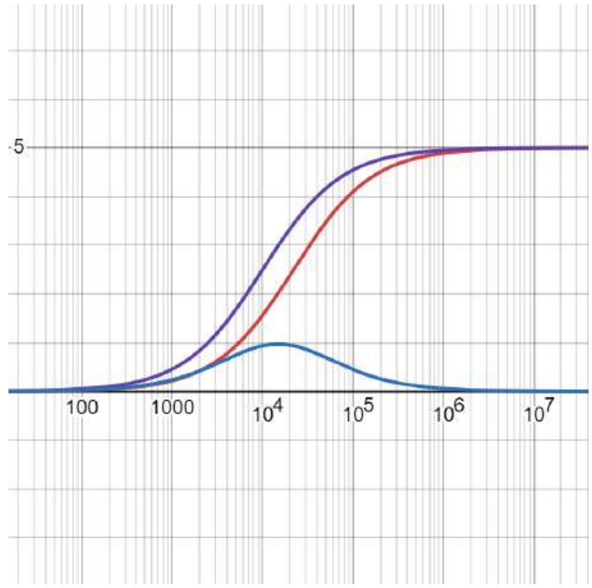

Figure 4.5: Graph of equations 3, 4, and 5 in Desmos.

This demonstrates that to maximize the sensitivity of the Arduino Nano's analog pin, R_1's resistance should be around 14kΩ, which yields a difference of nearly 1 volts. As 14kΩ and 15kΩ resistors aren't quite common, a 10kΩ resistor (Figure 4.7) is used as a substitute due it its prevalent use while also yielding a difference in potential difference of 0.938 (Figure 4.6), a result that is quite similar to the optimal 0.973.

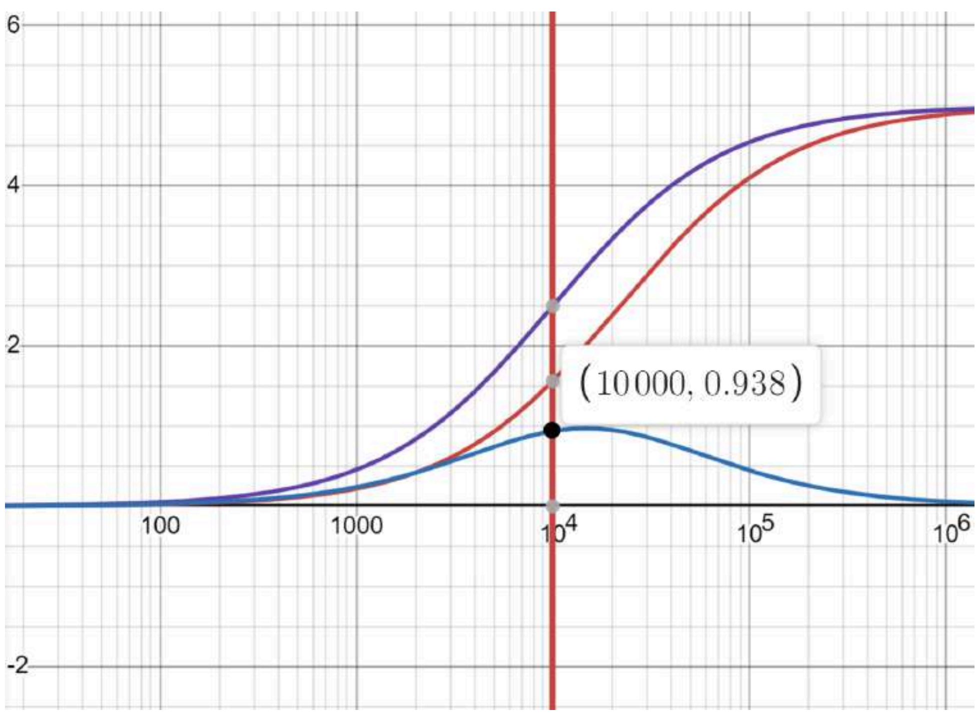

Figure 4.6: Graph of equations 3, 4, and 5 in Desmos showing the point on equation 5 with coordinates of (10000, 0.938).

Figure 4.7: 10 kΩ Resistor [1].

E. Servo Drive Debug Board

A servo drive debug board (Figure 4.8) is used to allow the program on the arduino board to control the degree of servo rotation and enables the connection of a detachable battery. This provides a convenient and intuitive linkage between the microcontroller board, battery and motors while also being quite compact.

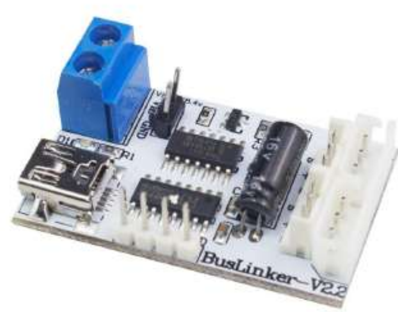

Figure 4.8: Bus Servo Driver Board [22].

F. Yahboom K210 Visual Recognition Module

The Yahboom K210 is used as the image processing hardware for this project. The K210 Visual Recognition Module allows for image recognition deep learning algorithms to be imported, an essential component that allows for a seamless integration of the automatic object grab functionality into the user's daily life. Additionally, its small size of 41.21mm x 57.27mm and lightweight design of only 130.4 grams increases its portability and allows the user to perform a range of rehabilitation exercises in a state of physical ease without exerting excessive effort in resisting the weight of the sensor.

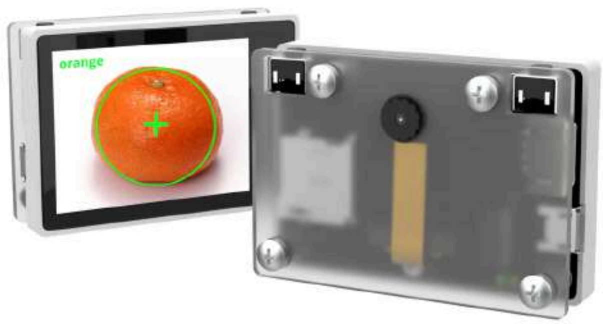

Figure 4.9: Yahboom K210 Visual Recognition Module [24].

G. Ultrasonic Sensor HC-SR04

To aid the Yahboom visual recognition module in evaluating the distance of the object relative to the hand, the ultrasonic sensor HC-SR04 was chosen due to its high accuracy and precision in measuring distance. This sensor emits a high-frequency sound through its trig pin which reaches the object. Afterwards, the wave is reflected back from the object to the echo pin, an ultrasonic receiver which calculates the distance of the object from the sensor through the time between sending and receiving the ultrasonic wave.

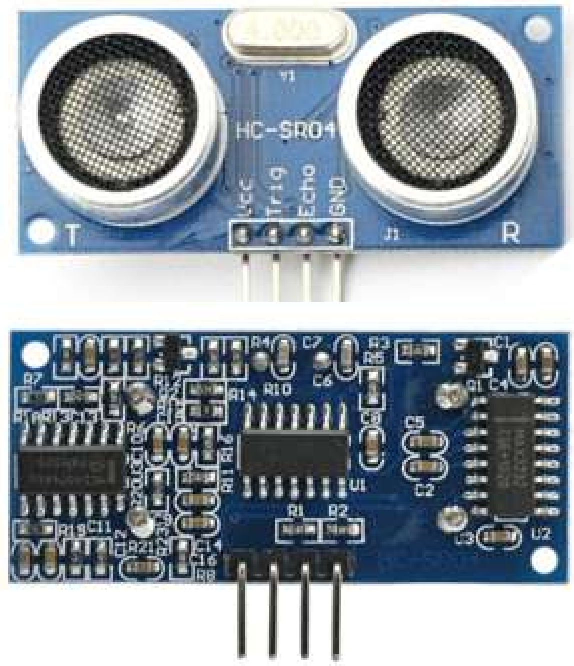

Figure 4.10: Ultrasonic Sensor HC-SR04 [5].

H. Circuit diagram

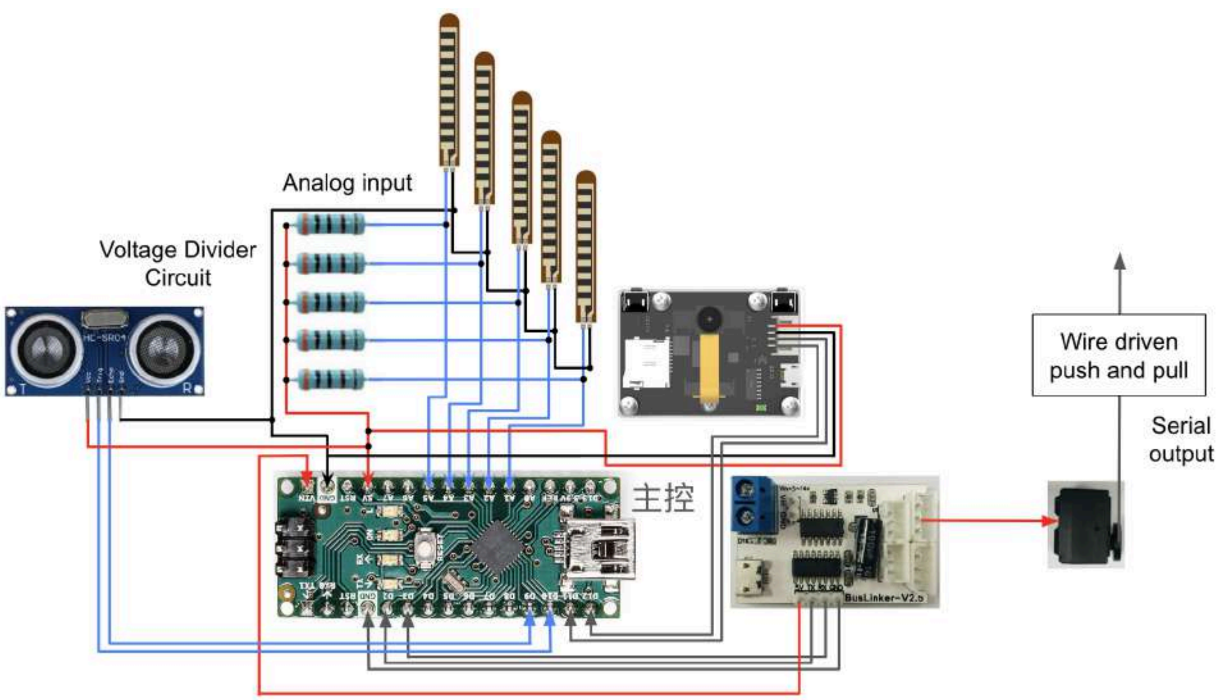

Figure 4.11: Electronic wiring diagram of the rehabilitation glove.

This system utilizes an analog pin from the Arduino Nano which detects the change in voltage in the 5 voltage divider circuits consisting of a 10kΩ resistor, a flex sensor and an Arduino Nano analog pin. The pin reads the voltage of the circuit, ranging from zero to five volts, of the 10kΩ resistor and converts it into a numerical value ranging from zero to 1023. This numerical value corresponds to a power of two within the comparator. As the the 10kΩ resistor's voltage measured corresponds to the flex sensor's arc angle, which subsequently correlates to the flexion and extension of the finger, this allows for each finger's degree of flexion to be mirrored to the other hand through motor rotation in Mirror mode.

4. Program Design

1) Manual Exoskeleton Movement Program

The flow chart of the program, written in Arduino IDE, is as follows:

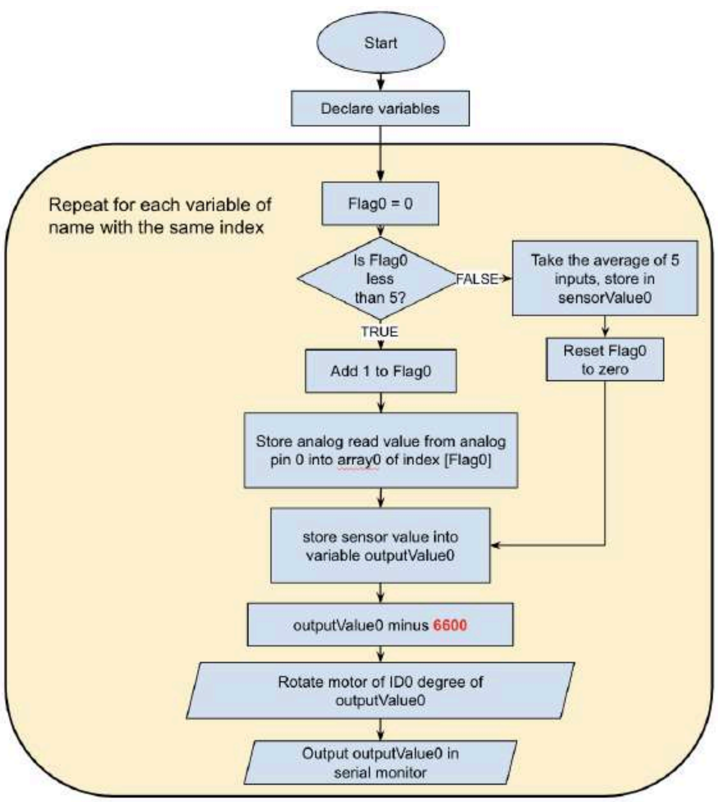

Figure 4.12: Flow chart of the program written in Arduino IDE.

The above flow chart only illustrates the logic behind the program for the index finger and iterates the same code, with some minor variable changes, for the other 4 fingers to minimize space. For the other 4 fingers, variables that end with a number, such as “Flag0”, will end with other numbers corresponding to their finger index number (Table 1).

Table 1: Finger index and its corresponding buffer number.

Finger	Index number	Buffer Number
Index	0	6600
Middle	1	6600
Ring	2	6900
Pinky	3	6600
Thumb	4	6300

For instance, the variable“Flag0” for the index finger will be replaced with “Flag3” for the pinky finger, as the pinky finger has an index number of 3. Additionally, a buffer number, marked in red in Figure 5.12, is added to allow for a more precise mapping of sensorValue to outputValue, their ranges being 0-1000 and 0-240 respectively.

2) *Hand Comfort State Identification Program*

To accurately assess the comfort level and status of rehabilitation, an image recognition and detection model, trained through the platform of MaixHub, was utilized. The model was trained based on 120 images of the researcher’s hand wearing the rehabilitation exoskeleton glove performing each of the 3 different hand positions behind a white background: Comfort, Discomfort and Extreme Discomfort (Figure 4.13). The model categorizes distinct hand positions based on individual digits’ abduction or adduction degree and their flexion and extension degree. For comfort, the patient’s individual fingers are spread evenly apart and the thumb is on the same plane as the other four fingers. Discomfort is categorized as the thumb experiencing slight flexion and the other four fingers touching each other tightly on the same plane. As for Extreme Discomfort, all four fingers and the thumb are in close contact with each other as shown in Figure 4.13.

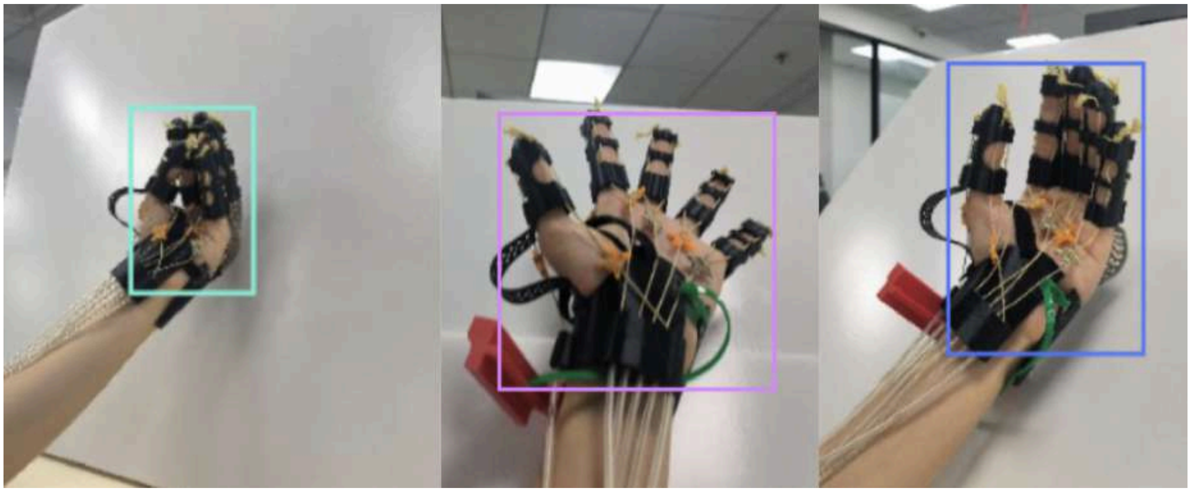

Figure 4.13: The three hand positions trained by image model: Comfort, Discomfort and Extreme Discomfort.

These images of the 3 different hand poses were taken from a variety of angles and positions to increase the model's accuracy in detecting the correct hand position (Figure 4.14). However, as the hand position of Comfort is similar to that of Discomfort (Figure 4.15), the researcher imported another 60 images of the hand position for Discomfort, accumulating a total of 480 images in the model's training dataset.

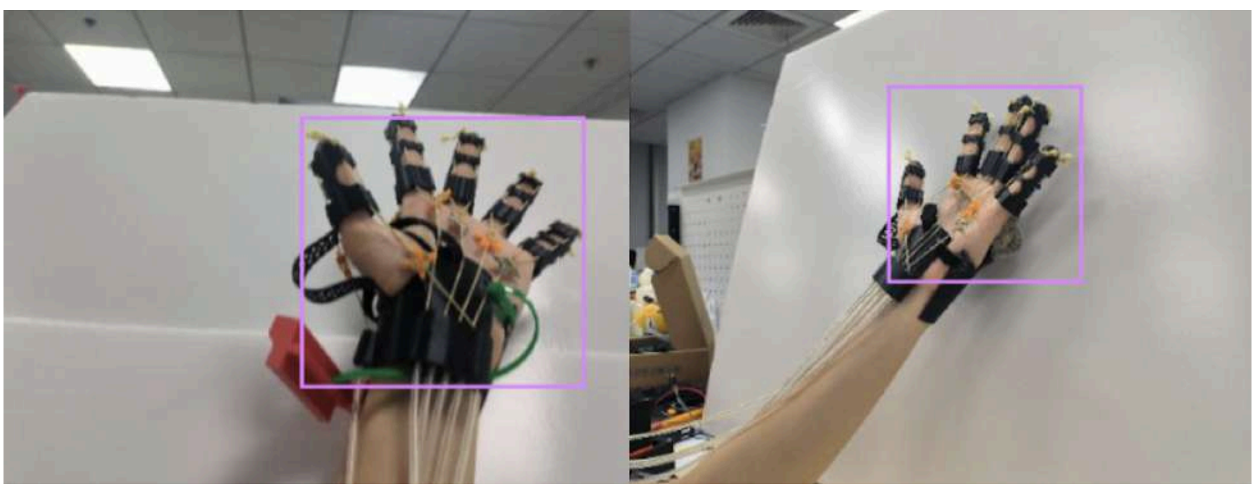

Figure 4.14: Images of Comfort hand position taken from multiple angles.

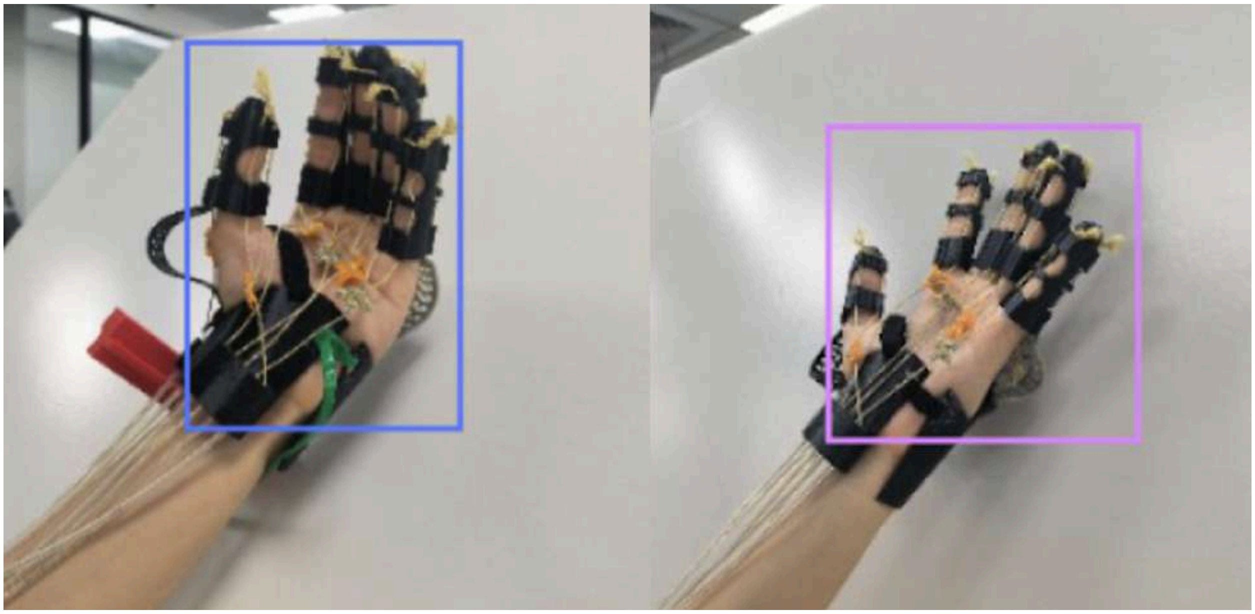

Figure 4.15: Image illustrating the similarity between Comfort and Discomfort hand position.

The principle behind the identification of hand position is as follows: The model aims to capture the patient's ability to perform the task of spreading their palm wide, and rates their comfort level on a three point scale. As patients with hand motor function impairment experience what is known as hypertonia, the muscle tightness and thus difficulty in opening and closing their hands, this measuring scale is particularly effective at determining a patient's rehabilitation stage, allowing for the quantification and assessments of any improved range of motion or flexibilities made.

3) Automatic Exoskeleton Grab Program

To encourage stroke patients to use the device more frequently outside of clinical settings and in daily life, a program was developed to allow the user to independently complete functional tasks of daily living. Such activities include the manipulation of common household items such as bottles, cups and remote controllers using the exoskeleton glove, allowing for active assisted recovery through naturalistic grasping/manipulation motions. Using Canaan Technology's image training model, a dataset of 700 total images was imported. These images include some of the most common household items: 300 images of water bottles, 100 images of remote controllers, 100 images of apples, and 200 images of cups. These images were taken from a variety of angles and settings (Figure 4.16). Afterwards, bounding boxes were manually drawn around each item to label them. This allows the model to develop a more robust model by having a diverse dataset with a variety of images from multiple perspectives to train from.

Afterwards, the images were fed into Canaan Technology's image detection model for training. The model utilizes the K210 chip during training with a learning rate of 0.001. After 6 hours and 700 iterations of training, the final model achieved a detection accuracy of 99.9% (figure 4.17), indicating that the model is able to generalize beyond the given dataset and reliably recognise water bottles, remote controllers, apples and cups in a variety of settings, lightings and perspectives to an exceptionally high degree. This image recognition model was uploaded to the Yahboom K210 Visual Recognition Module through its memory card, with the module soldered and wired to the arduino board.

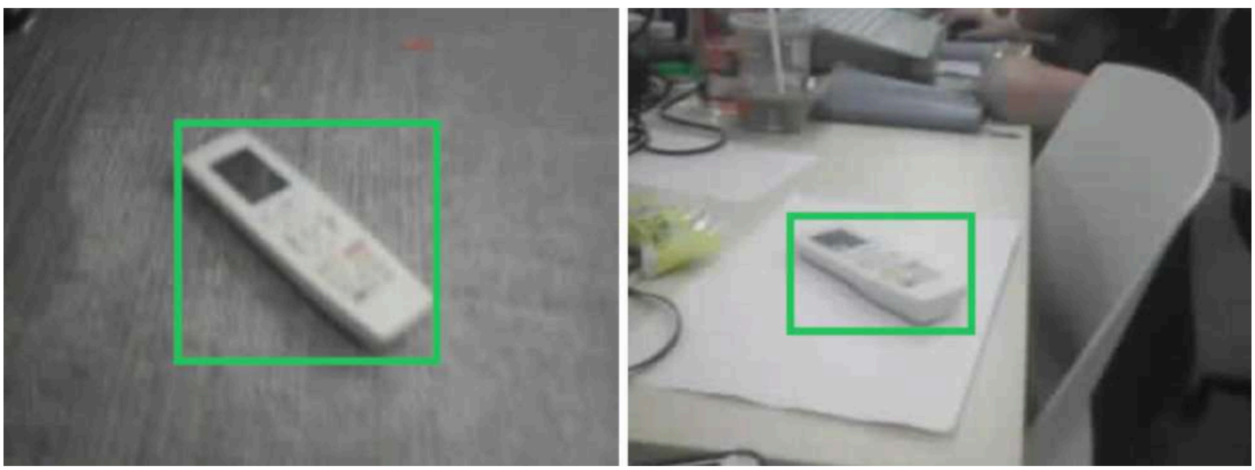

Figure 4.16: Images of a white remote controller taken from a gray floor and a white table top background.

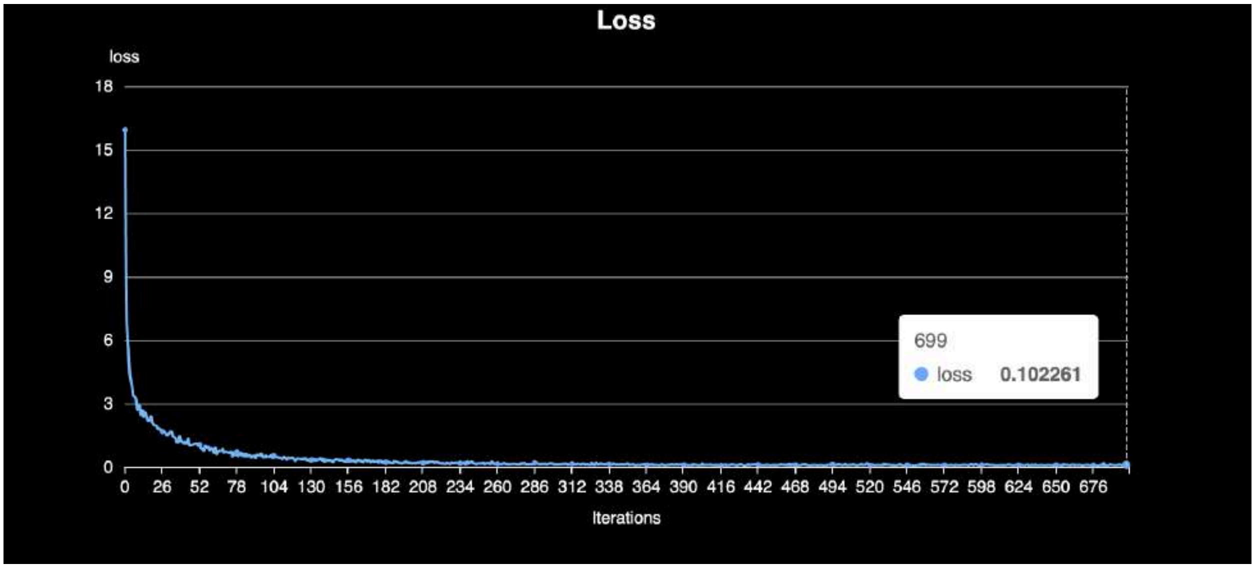

Figure 4.17: The difference between the predicted value of the model and the actual value of the dataset.

The current python code on the Yahboom K210 module outputs the width, height, label, accuracy score and the xy position of the detected object. However, only the label is useful in classifying the object as the distance can be measured more accurately through the ultrasonic sensor. The flow chart for the fully automated exoskeleton grab program shown below (Figure 4.18):

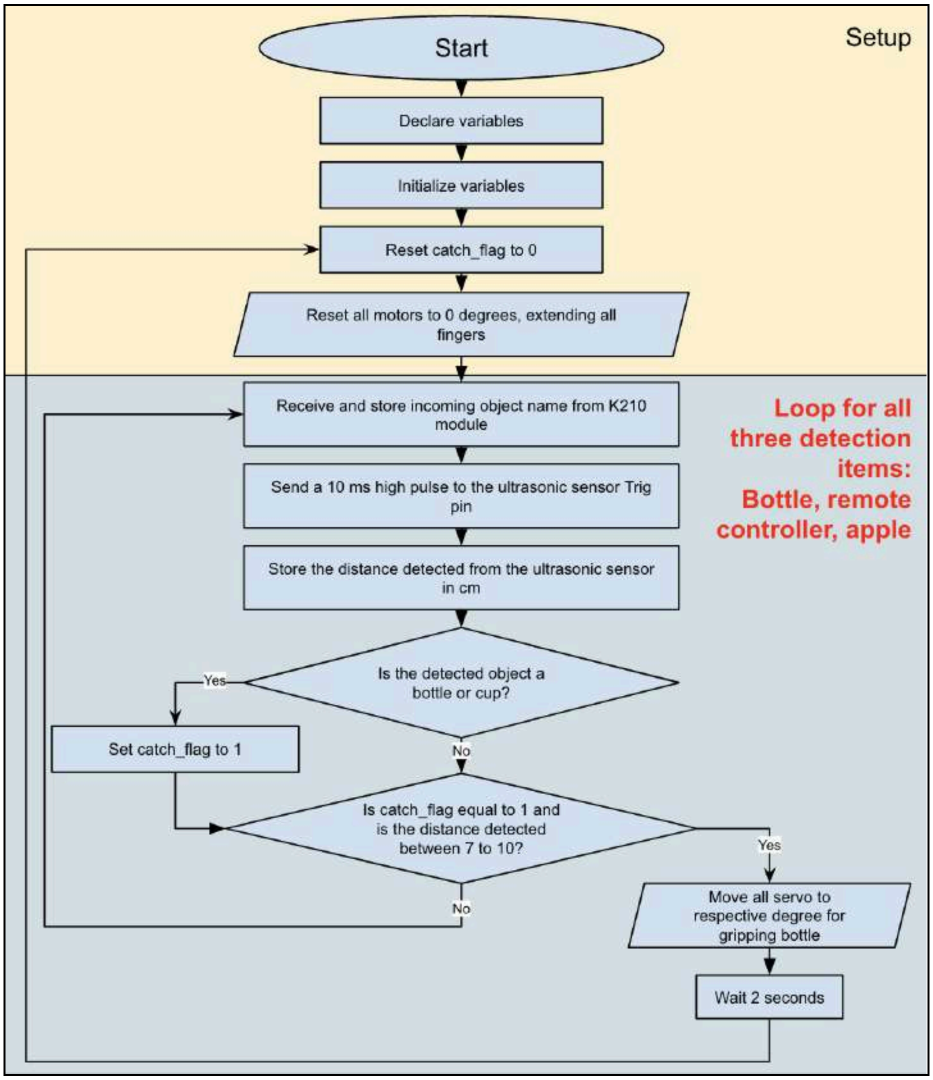

Figure 4.18: The flow chart of the Automatic Exoskeleton Grab Program.

III. Experiment

Experiment 1: Comparing Joint Flexion Resistance in Various Iterations of Exoskeleton Glove Finger Models

a) Introduction

The finger model has been modified multiple times by the researcher to improve comfort, add functionality and correct design errors. However, it has been put into question whether the adjustments actually made the finger model easier to move. Thus, the researcher set up an experiment to identify the effect of the width of a joint on the force needed to move that joint.

b) Material list

- 6 3D printed finger model made of TPU (Version 1, 2, 3, 4, 5, 6)
- 14mm of string
- A Spring-loaded thrust meter

c) Procedure

1. Cut 14 mm of string.
2. Thread the line through the front line holes of the finger model (version 1).
3. Tie the two ends of the line together.
4. Hook the line onto the hook of the spring-loaded thrust meter.
5. Hold the finger model firmly and pull the thrust meter slowly away from it until the first joint hole has "closed".
6. Record the measurement shown on the thrust meter appropriately.
7. Repeat steps 5 and 6 with the second joint hole.
8. Repeat steps 5 and 7 with version 2, 3, 4, 5, 6 of the finger exoskeleton model.

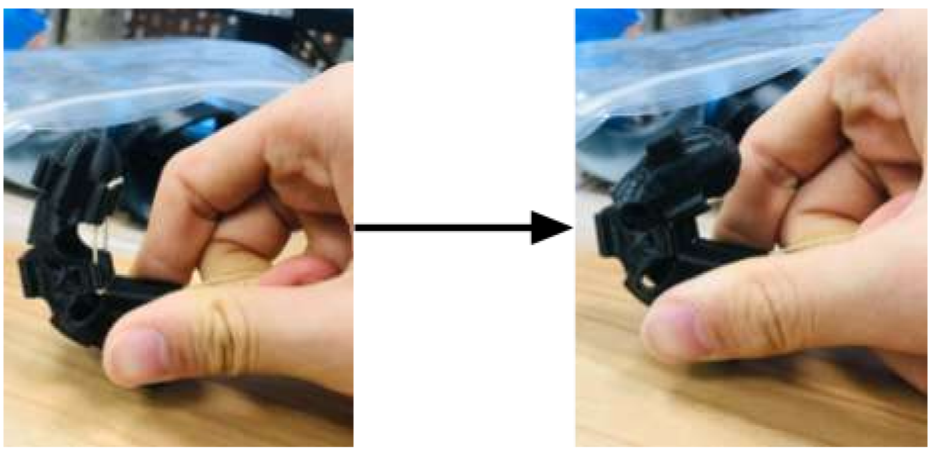

Figure 5.1.1: Fully "closed" version 6 of finger exoskeleton model.

Figure.5.1.2: Experimental setup of experiment 1.

d) Data Table

Table 2: Comparison between finger exoskeleton version, its characteristics and joint forces.

Photo & Color	Version	1st Joint force (N)	2nd Joint force (N)	Total force required (N)	Finger hole thickness (mm)	First joint side Width	Second joint side width	Guard flap thickness	Notes
White 	1	22.5	17.5	40	2	7.5	7	N/A	
Grey 	2	12.5	17.5	30	1.5	7	7	N/A	Printing defects, larger string holes
Grey 	3	9	26	35	1.5	6	7	N/A	Printing defects, longer joint cutout holes
Black 	4	7.5	5	12.5	1.5	6	5.5	1.5	Printing defect, thin guard flap weakly attached, longer joint cutout holes

Grey 	5	22.5	10	12.5	1.5	7*	7.5*	4	*Side width includes thick guard flap Printing defect, cut on edge of second joint cutout
Black 	6	7.5	7.5	15	1.5	6.5	6	1.2	Shortened guard flap width and adjusted angle of guard flap, when doing first test, second joint moved too

e) Results and Data Analysis

Figure 5.1.3 showed a weak positive correlation between the force needed to bend the first joint and the width of the first joint, with the exception of version 5. This correlation is evident between version 1 and 2, where the first joint side width decreased from 7.5 to 7mm which consequently caused a similar reduction from 22.5N to 12.5N in the force required to move the first joint.

However, this trend may not be so apparent between versions 3 and 4, which had the same first joint side width but a noticeable reduction in the force required to move the first joint from version 3 to 4. This may be due to the first joint side width not being the only factor that influences the force. For instance, the quality of 3D printing may also have a large influence over the force required to actuate joints. As version 4 had a couple of printing defects, such as weakly attached guard flaps, its total force required to move both joints drastically shifted from 35N of version 3 to 12.5N of version 4, a force that was significantly smaller compared to the previous iteration. However, similar forces may not correlate to similar first joint side widths either. This is shown in a comparison between iterations 3 and 6, where although version 3's first joint side width was 0.5 mm smaller than that of version 6's, the finger exoskeleton still required 1.5N more force to move the first joint. Similar to the comparison between versions 3 and 4, versions 3 and 6 had other factors that influenced this anomaly, which in this case was a shortened guard flap width and an adjusted attachment angle of the flap.

Thus, this experiment does not identify one characteristic of the finger exoskeleton that leads to the least force required to actuate it, but rather a combination of different factors. Version 6 strongly indicates this. Through a combination of large joint cutout holes, thin, short and angled guard flaps, and similar forces required to actuate both joints, version 6 improves comfort and allows a natural movement of the finger.

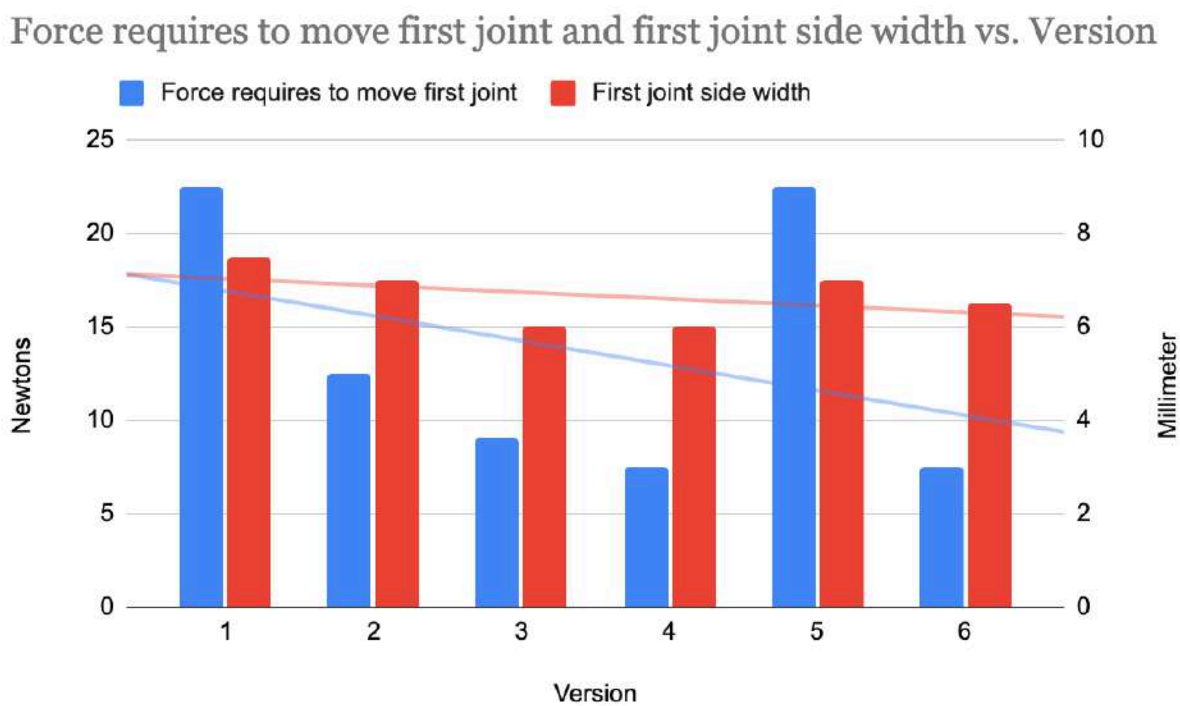

Figure 5.1.3: Graph comparing force required to move first joint and side width of first joint vs. version.

f) Conclusion

Through this experiment, the researcher determined the relationship between design characteristics of a finger exoskeleton and the actuation force required to move both joints. While the results showed a general trend of decreasing joint width that corresponds to decreased actuation force, the experiment aims to also highlight the influence of other design factors. Version 6 best demonstrates this by embodying a balanced set of optimized design properties, including large cutouts, thin angled flaps, and balanced joint forces, and is thus chosen as the final iteration design.

Experiment 2: The influence of different filtering algorithms on the vibration of a motor in a stationary state.

a) Introduction

After running the program, the researcher noticed that even when the flex sensors were stationary, slight motor rotations, in the form of vibrations, still occurred. After further inspection, it was concluded that due to the nature of the program, subtle changes of the flex sensor were mirrored to the motors, causing unwanted vibrations shown in Figure 5.2.1.

Figure 5.2.1: Graph of motor vibration plotted in serial monitor of arduino IDE.

Thus, a second program was proposed. Program 2 takes 5 samples of readings from the flex sensors, and transfers an average of the 5 data points to the motor. This method aims to reduce the unintentional movement of flex sensors as the motors rotate every 0.5 seconds instead of the previous 0.1 seconds, diminishing unwanted vibrations.

To measure the vibrations, a camera captures the movement of a white ball, attached to a wooden stick that extends from the specialized servo wheel which is connected to the servo. This allows for the quantification of servo vibration, which translates into the movement of the white ball that can be tracked using a camera and analyzed through applications such as Tracker. Using Tracker, the changes in the x-position of the white ball was recorded in a data table and plotted against time (Figure 5.2.4).

b) Material List

- Wooden Stick (20 cm long, 0.3 cm diameter)
- White ball (~0.8 cm diameter)
- Glue gun
- Specialized servo wheel
- Two 6 mm M2 screws
- M2 screwdriver
- Glove with flex sensor attached
- Motor box
- Computer with the arduino programs one and two (Types: Original, average)

c) Procedure

1. Glue the white ball onto the end of the wooden stick with a hot glue gun.
2. Attach the stick into the specialized servo wheel securely.
3. Screw two 6mm M2 screws through the specialized servo wheel and the original servo wheel using a M2 screwdriver. The wooden stick should be pointing vertically upwards.
4. Attach original servo wheel onto servo.
5. Connect motor to motor box.
6. Tape motor onto the table.
7. Connect Arduino Nano to the computer.
8. Place a camera aligned but about 7 cm from the white ball.
9. Wear the opposition hand glove on left hand and place it on a flat surface within the camera's range, with all fingers extended (Figure 5.2.2).
10. Run program 1 on the computer.
11. Press record on the camera, ensure that the white ball is within the camera's range.
12. Wait 1 minute.
13. Stop recording on the camera.
14. Upload file onto the Tracker application.
15. Press axis and select the white ball as origin.
16. Press Track, "new", and select "Point Mass".
17. Press Calibration, "new", and select "Calibration Stick".

18. Drag one end of the calibration stick onto one side of the white ball, and the other end onto the other side of the ball.
19. Customize the length as 1.400E-2 m.
20. Shift control click the center of the white ball. This creates a keyframe.
21. Press Mass A at the top left and click Autotrack.
22. Press search and wait for the process to complete.
23. After the process is complete, select “file” on the top left, “export”, and “data”.
24. In the Tracks tab, select “mass A”. In the Columns tab, select “t, x”.
25. Press Save As, and click txt file.
26. Repeat steps 9 to 25 with hands in a fist using program 1, and with hands extended and in a fist using program 2.

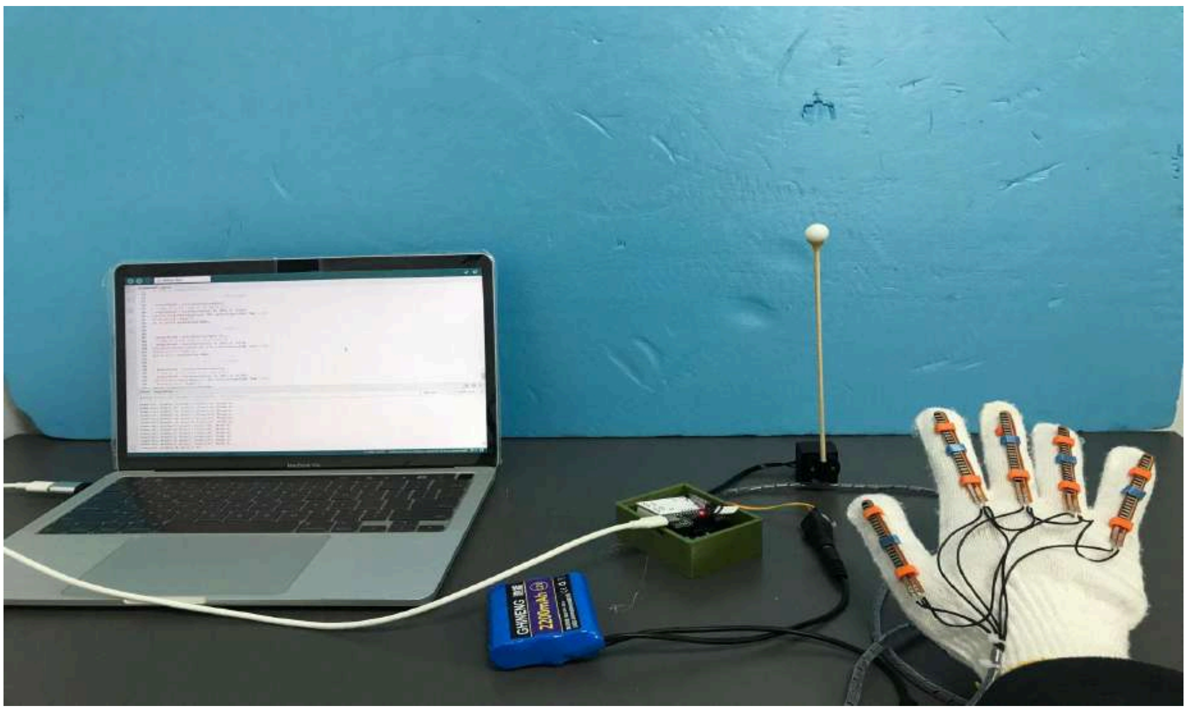

Figure 5.2.2 Experimental Setup of experiment 2.

d) Data Table

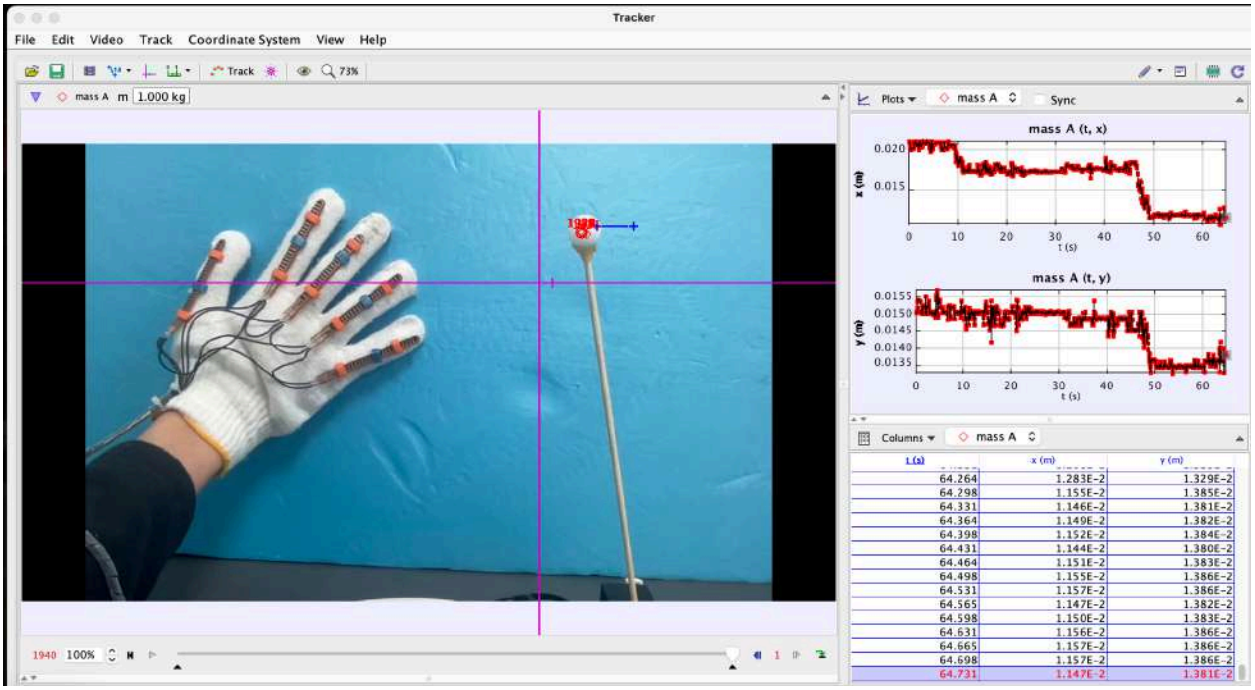

Figure 5.2.3: Experiment display in Tracker application.

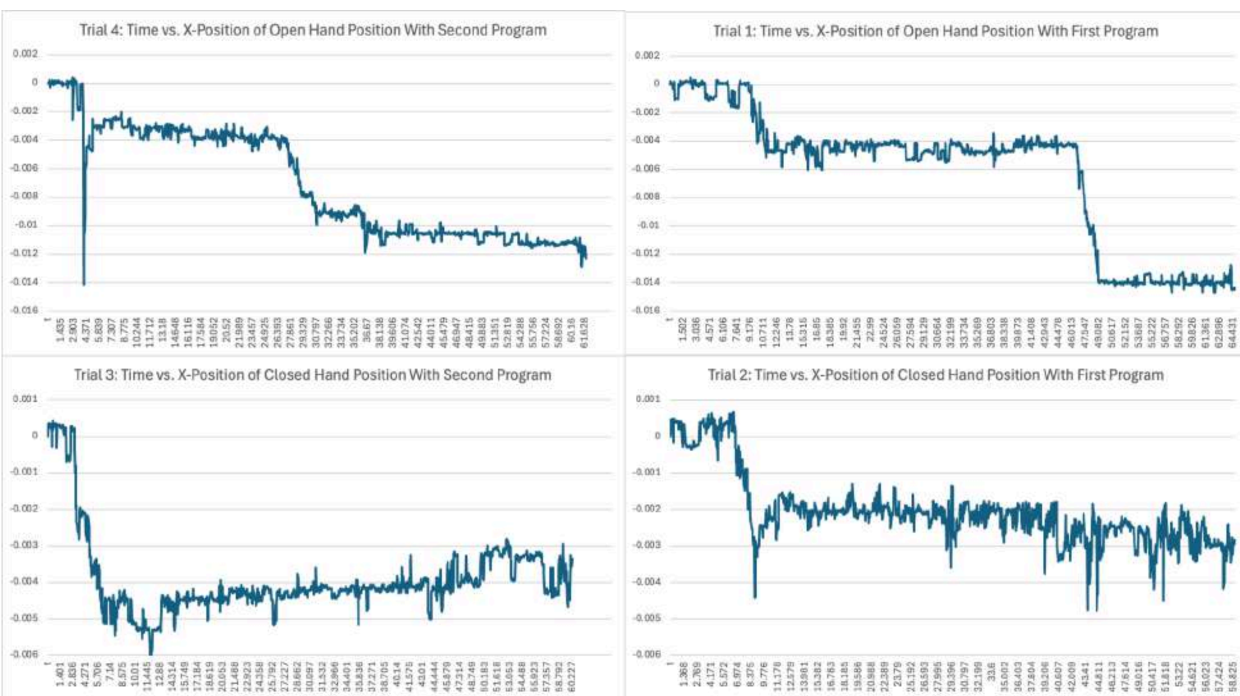

Figure 5.2.4: Graph of Time (Seconds) vs X-Position for the 4 Trials

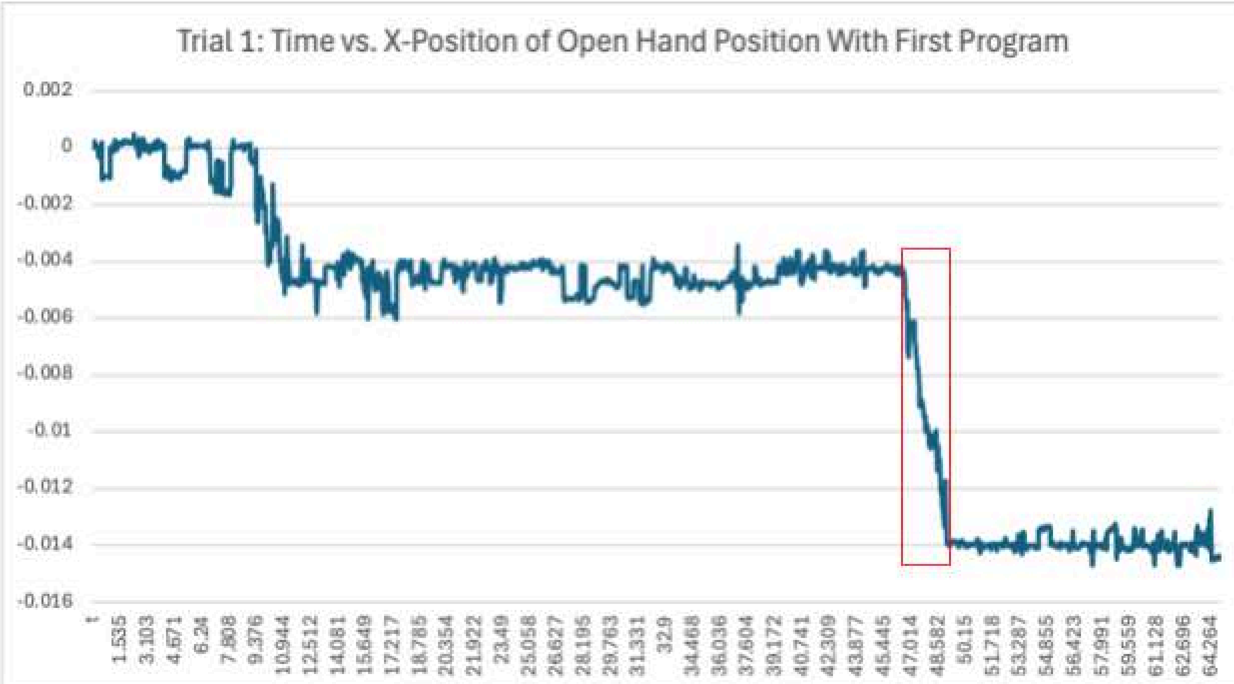

Figure 5.2.5: Graph of Time (Seconds) vs X-Position for Trial 1 Highlighting t=46s to t=49s.

e) Results and Data Analysis

The exported data from Tracker was imported into Microsoft Excel for data analysis, with Figure 5.2.4 showing the 4 graphs of the variations in the two programs along with the two different hand positions respectively. Upon inspection of trials 1-4, it was noticed that all of the above graphs contained an anomaly of a sudden drop in the x position that is significantly larger than the other changes in the x position of the same graph within a short period of time, an example being t=46s to t=49s in the graph of trial 1(Figure 5.2.5). As these sudden large magnitude changes do not reflect the slight motor movement vibrations investigated in this experiment but is instead due to unintended movements by the researcher, to derive meaningful calculations and avoid these anomalies, the researcher has decided to only investigate and analyze the longest continuous period where the x-position remains relatively stable while ignoring those sudden fluctuations. This period of time is abbreviated as **SP** (which stands for Selected Period). The SP for each trial is shown in Table 3.

Table 3: Trial name and its corresponding SP range.

Trial Name	Trial #	SP range (s)
Open Hand Position With First Program	1	9.343 - 46.647

Closed Hand Position With First Program	2	9.91 - 61.795
Closed Hand Position With Second Program	3	13.08 - 60.227
Open Hand Position With Second Program	4	4.404 - 27.561

Additionally, to quantify and convert these momentary vibrations of a small magnitude to meaningful data, the standard deviation was calculated for the SP of each trial. A low standard deviation indicates that the values tend to be closer to the mean than trials with a high standard deviation, suggesting that the vibrations have a lower magnitude and thus indicating less variation and vibration during the motor's movement. Thus, the standard deviation is a compelling measure of a program's effectiveness at reducing redundant vibrations. The standard deviations of each trial's SP, expressed as s, is calculated below:

$$s = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n-1}}$$

Where:

x_i = Specific value in the sample

$\bar{x}$ = Sample mean

$(x_i - \bar{x})$ = The distance the specific value in the sample is from the mean

n = Sample size

Using Excel's built-in function =STDEV, the standard deviation value of each trial is shown in Table 4.

Table 4: Trial name and its corresponding standard deviation range.

Trial Name	Trial #	Standard Deviation Value
Open Hand Position With First Program	1	0.000580892
Closed Hand Position With First Program	2	0.003290096
Closed Hand Position With Second Program	3	0.000437891
Open Hand Position With Second Program	4	0.000527915

The standard deviation value for trial 4, 0.000528 is less than that of trial 1, which has a SD value of 0.000581. The difference of 0.0000530, although a small difference, suggests the second program is more effective at reducing vibration than the first program. As program two takes the average of 5 readings from the flex sensors to send to the servos rather than sending each reading to the servos, the servo vibration would not only occur less frequently, it would also diminish the magnitude of vibrations as subtle movements of the flex sensors would be diluted with other flex sensor readings. This elucidation of the effectiveness of program two is further supported by the comparison between trial 2 and 3, which compares the standard deviation values of the two programs in a closed hand position. The utilization of the second program, shown in trial 3, achieved a lower SD value than that of trial 2 which employed the first program by a significant margin of nearly 80%, further reinforcing program two's effectiveness.

f) Conclusion

When comparing the standard deviation values of a ball extended from a specialized servo wheel's x-position during servo vibrations, program two had a lower standard deviation value when compared to program one. Thus, program 2's algorithm is the most effective at reducing redundant vibrations among the two testing algorithms.

Experiment 3: Comparing Different String Length Adjusters' effectiveness and stability

a) Introduction

As mentioned previously in the Structural Design section, the wire-driven system of the hand exoskeleton glove requires the wires to be completely straight in order to provide tension force that causes flexion of the finger. Therefore, string length adjusters were created to maintain the string tension while minimizing space and maximizing comfort. Since the pulling force of the servo motors on 11.1 volts is given as 12kg/cm, the maximum force exerted on the string can be expressed as 117.72 Newtons. Although the theoretical maximum force the motor can exert is more than 100 newtons, a more reasonable testing range for this experiment would be 0-50 N, as running the motor at their maximum speed would not reflect the intended rehabilitation usage. Two criterias were created to measure the effectiveness of the string length adjusters: Critical force and one rotation length. The critical force is a measure of the maximum amount of force exerted on the string length adjusters before deformation. The second criteria, one rotation length, is the measure of the length of string required to make one full loop around the string length adjusters.

b) Material List

- 3 String length adjusters (Version 1, 2, 3)
- A Spring-loaded thrust meter
- A vernier caliper

- 200 mm of string

c) Procedure

1. Measure and cut 200mm of string.
2. Tie one end of the string to the spring-loaded thrust meter.
3. Measure 100 mm of rope using the vernier caliper and mark both ends of the 100 mm appropriately.
4. Hold both markings together, forming an "Ω"-like shape.
5. Poke the tip of the folded portion of the string through the hole in the version 1 of the string tyre.
6. Twine the string around the horizontal columns, referred to as legs in an alternating up and down fashion.
7. When the string is unable to reach the next leg, hook the string around the nearest leg.
8. Reset and adjust the thrust meter until it shows 0 N.
9. Pull onto the other end of the string and incrementally increase the force until any noticeable deformations occur in the string length adjuster.
10. Once any deformation occurs, record down the force, referred to as the critical force required to cause the deformation appropriately.
11. Unwrap and remove the string from the string string length adjusters.
12. Using one marked end of the 100 mm string, press against a point on the side of the string length adjuster.
13. Loop the string around the string length adjuster until it has reached the marked end.
14. Mark the point where the marked end of the string and the looped string overlapped.
15. Unwrap and remove the string from the string length adjuster.
16. Measure using a vernier caliper the distance between the marked end of the rope used in step 12 and the newly marked point. This distance is labeled as "one rotation length".
17. Repeat steps 4 to 16 with string length adjusters version 2 and 3.

d) Data Table

Table 5: Iteration of string length adjuster and its critical force performance.

Version names of string length adjusters	Critical Force (N)	Changes	One rotation length

1 - Cylinder 	22	Arms were damaged easily and severely, causing a lot of string to release tension	33
2 - Octopus 	24	Slight staggering of arms and cracks forming in the center circular region	39
3 - Cross 	38	Slight staggering of arms	40

e) Results and Data Analysis

In this comparison between string length adjusters, 3D printed parts that maintained tension between the finger exoskeleton and the motor, the Cylinder string length adjuster had the least critical force of 22N. This may be due to the Cylinder's weak arms that are only a few millimeters wide, allowing the strings to easily deform its shape under slight tension. Thus, a disadvantage of this design is the large scale deformation of arms under tension. Additionally, its one rotation length is also significantly shorter than that of version 2 and 3. As the Cylinder has to stay within the dimensions of the metacarpophalangeal (MCP) joint, its structural limitations restrict its radius, leading to a diminished distance of its one rotation length.

The next iteration, the “Octopus”, had a notable improvement in its one rotation length due to the interlacing of the string with its arms and the string stacking structure (Figure 5.3.1) as the string goes in a loop, increasing its string capacity. There is also a slight increase in the maximum amount of force tolerated, from 22N to 24N. This can be attributed to the increased width of the arms, leading to the higher tolerance of stress from the string.

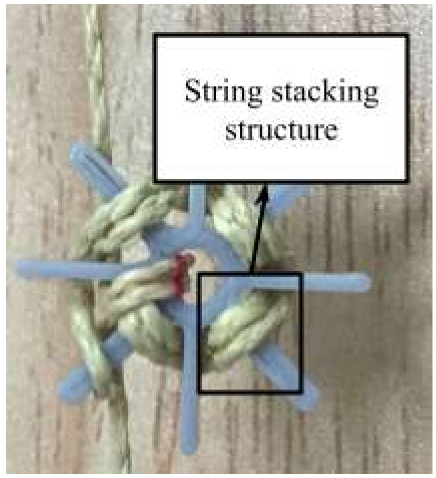

Figure 5.3.1: String stacking structure of the “Octopus” string length adjuster.

The “Cross” out performed the other two string length adjusters in terms of both critical force and rotation length (Figure 5.3.2), withstanding 14N more force than the previous iteration while also having a slightly longer rotation length. The exceptional surge of the critical force is as a result of the new design that closely spaced pairs of arms together while creating new curvatures between the pairs, enlarging the base of the string length adjuster and thus increasing the force required for fracture.

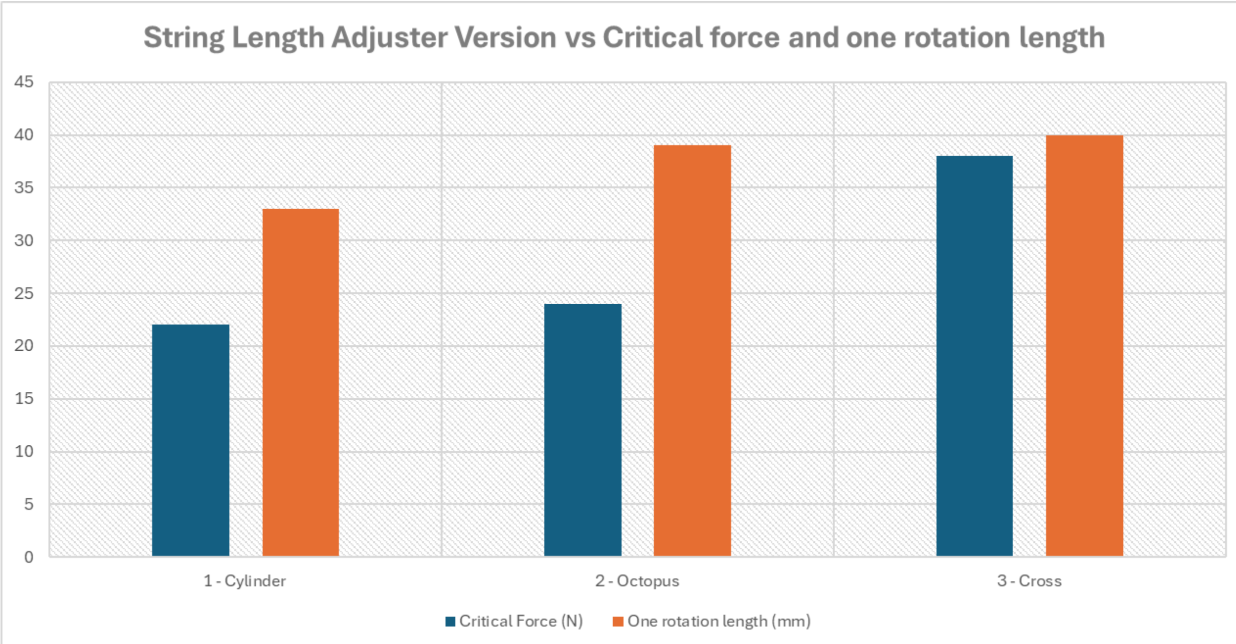

Figure 5.3.2: Graph comparing version of string length adjuster to its critical force and one rotation length.

f) Conclusion

The results of this experiment, an evaluation of 3 types of string length adjusters, showed improvements in the critical force and one rotation length with each iteration. Comparing each version, iteration 3 had both a higher critical force and a longer one rotation length due to its closely spaced arm pairs to widen the base and create new curvatures, substantially increasing the critical force tolerated. Therefore the “Cross” structure is chosen as the most optimal design for a string length adjuster.

Experiment 4: Force Exerted by Motor During Finger Flexion

a) Introduction

The loss of the ability to extend individual fingers of the hand and thumb has been frequently observed in stroke patients. They typically experience a clenched fist or a tense thumb. To measure the effectiveness of the hand rehabilitation glove in alleviating these motor impairments, an efficacious metric would be the amount of force exerted by the glove during a complete flexion of individual digits of the hand.

b) Material List

- Rehabilitation device
    - Glove with flex sensor attached
    - Motor box
    - Exoskeleton Glove
- A Spring-loaded thrust meter
- 100mm of string
- Desk clamp

c) Procedure

1. Tie one end of the 100mm string to the other end to form a loop.
2. Clamp the desk clamp tightly onto the table. Ensure there's a stable handle available to lean hand on.
3. Wear an exoskeleton glove on left hand.
4. Reset the spring-loaded thrust meter to zero newtons.
5. Place the circular string around the pinky finger's first joint of the exoskeleton glove.
6. Place the other end of the string around the hook of the thrust meter. Ensure that the string maintains tension and the thrust meter still reads zero newtons.
7. Wear glove with flex sensor attached on right hand.
8. Lean the right side of the palm of the left hand onto the desk clamp handle.
9. Have a second researcher hold onto the spring-loaded thrust meter stably. The experimental setup should be akin to Figure 5.4.1.
10. Actuate the pinky finger using the movements from the motor by setting the motor degree to 240° from the arduino program. Ensure that the hand of both the exoskeleton glove and holding the thrust meter does not shift in its position by pressing against the handle of the desk clamp or against the table.
11. Wait 2 seconds.
12. Extend the finger by setting the motor degree to 0° using the arduino program.
13. Record the reading of the thrust meter.
14. Repeat steps 4 to 13 two more times with the pinky finger.
15. Repeat steps 4 to 14 with the index, middle, ring finger and the thumb respectively.

Figure 5.4.1: Experimental setup of experiment 4.

d) Data Table

Table 6: Finger versus force exerted during actuation.

Finger	Force (N)			
	Trial 1	Trial 2	Trial 3	Mean
Thumb	10.5	12.5	11.5	11.5
Index	12.5	10.5	12	12
Middle	11	12	12.5	12.5
Ring	10.5	10.5	14	12
Pinky	15	12	11.5	12.8

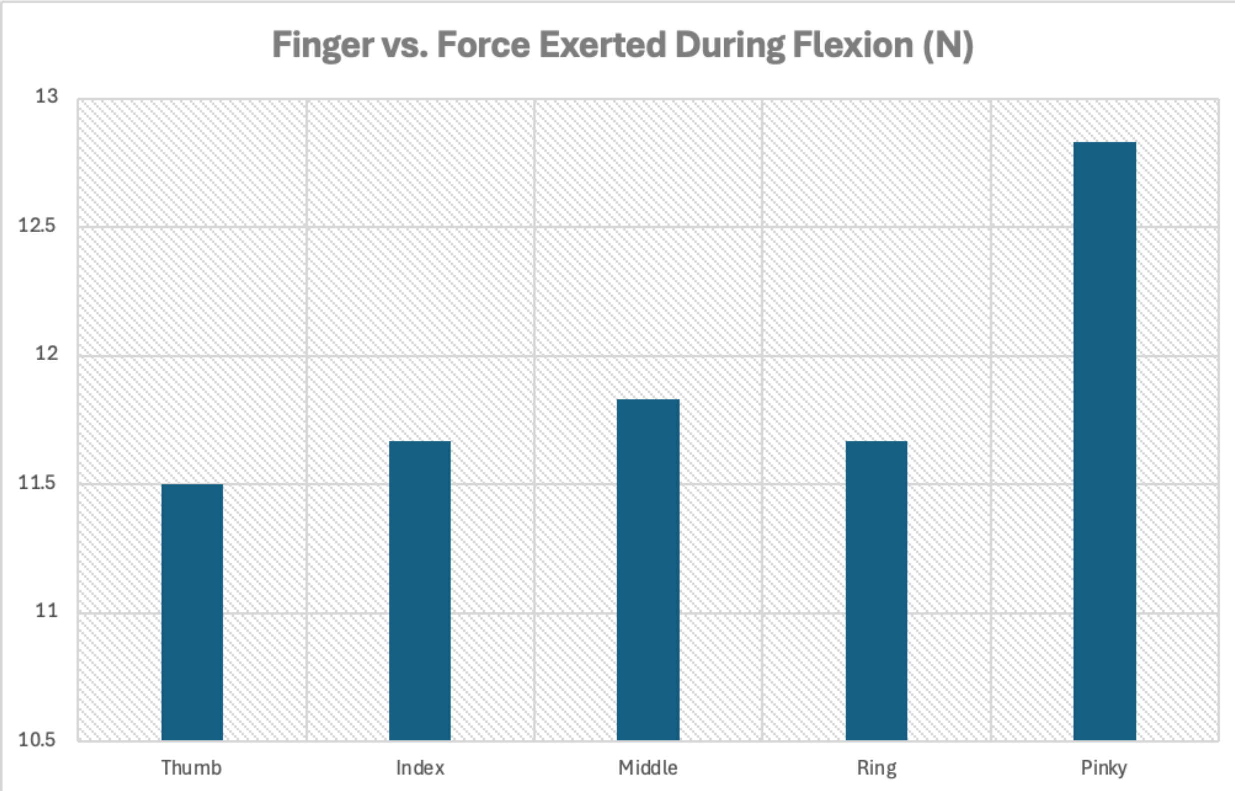

Figure 5.4.2: Graph comparing finger name to force exerted during finger flexion.

e) Results and Data Analysis

In this experiment, all of the digits of the finger displayed a force of 12±0.5N during flexion other than the pinky finger. This trend indicates a relatively equal distribution of force when grasping using the exoskeleton glove, suggesting that the glove is able to flex each finger with comparable levels of assistance. This reinforces the glove's effectiveness as a tool during active assisted hand recovery for the patients. The anomaly of the pinky's force being significantly higher than the 12N mean of the other digits, with a 0.8N increase of force is due to its small anatomy. When the same torque is exerted by the motor to rotate to 240°, the force is dissipated over the length of string that extends from the motor box, through the digit's exoskeleton to the tip of the finger. However, as the pinky has a shorter distance from the MCP joint to the tip of the finger relative to the other digits, its exoskeleton would also be shorter, corresponding to a reduced length of string. This signifies that the finger is able to dissipate less of the applied force over the shorter travel distances compared to the other four digits, which is demonstrated by the bar chart in Figure 5.4.2. A smaller finger width at the Proximal Interphalangeal and the Distal Interphalangeal joints, a significant marker in terms of the dimensions of the exoskeleton's joint openings will also lead to a greater proportion of the motor's torque directly transferring into the flexion force measurement by the thrust meter.

f) Conclusion

The experiment establishes the hand exoskeleton glove as effective due sufficient force exerted by the fingers, indicating its efficacious functionality during active assisted hand recovery. Additionally, the force ascertained of 12N is more than the required amount of force to securely grasp common household objects, enabling stroke patients to recover grasping capabilities for activities during daily activities involving grabbing motion.

Experiment 5: Accuracy Assessment of the Hand Comfort State Identification Program

a) Introduction

Due to the unreliable nature of AI image models, the accuracy of the Hand Comfort State Identification Program has been put to question. Thus, to verify the validity of the program, an experiment is conducted in an office setting to detect elements of improvement.

b) Material List

- Exoskeleton Glove
- Computer with web camera and recording software installed
- Ruler with 50 cm capacity

c) Procedure

1. Open a browser window and visit MaxiHub website.
2. Open a screen recording software on the computer and start recording.
3. Wear the exoskeleton glove on the left hand.
4. Lift the glove so that the entire exoskeleton glove and hand is within the main web camera frame at the center of the webpage. The arm should be perpendicular to the surface of the table.
5. Using a ruler, measure 50 cm from the base of the computer screen to the base of the left hand. Adjust the hand position accordingly until it's 50 cm away from the screen.
6. Open the hand wide, ensuring that the palm is flat and the fingers are on the same plane as the thumb. The palm should be parallel to the computer screen. Verify again that the glove is within the webcam's frame.
7. Wait 3 seconds.
8. In the same position, adduct the fingers together until the four fingers are touching each other. Move the thumb slightly forwards in a flexion motion by approximately 60°.

9. Wait 5 seconds.
10. In the same position, move the four fingers forwards by about 70°. Reposition the thumb until it's in contact with the index finger.
11. Wait 5 seconds.
12. Move the base of the arm forwards by 20 cm. Verify that the arm is perpendicular to the surface of the table.
13. Repeat steps 6 to 11.
14. rotate the palm of the hand ~30° to the left.
15. Repeat steps 6 to 11.
16. Save and quit the recording software.

d) Data Table

Table 7: Dynamic timestamps versus Hand State Program detection accuracy and angle of grab.

Trial #	Timestamp (s)	Hand State*	Angle (°)	Identified state*	Outcome	Notes
1	00:02	C	0	C	Success	
2	00:05	D	0	D	Success	
3	00:08	ED	0	ED	Success	
4	00:11	C	0	C	Success	
5	00:14	D	0	D	Success	
6	17-21	ED	0	ED	Success	
7	23-25	C	30	C	Success	
8	25-28	D	30	C and D	Mixed	Overlapping of the bounding box for “Discomfortable” and “Comfortable”
9	28-31	ED	30	ED	Success	

* To conserve space in this report, the following abbreviations are used: "C" to represent "Comfortable", "D" to represent "Discomfortable", and "ED" to represent "Extremely Discomfortable".

e) Results and Data Analysis

The above Hand State Recognition Model demonstrated a very strong accuracy in detecting the correct hand state, correctly identifying the hand state in 8 out of the 9 total trials shown in Table 7. Even with a variety of hand orientations and distance away from the camera, the camera was still able to achieve an 89% accuracy, illustrating the model's reliability and robustness in detecting small changes in the hand position. However, this system still has a large area of improvement. In trial 8, the only trial marked as "Mixed", even though the model correctly marked the hand gesture as "Discomfortable" with an 81.24% confidence score, it marked the lower half of the hand as "Comfortable" with a lower 52.86% rating (Figure 5.5.2). This ambiguous case illustrates the model's uncertainty in marking hand gestures shown at an angle, in this case rotated 30° to the left, indicating a need for a longer model training time and a larger training dataset regarding intermediate hand positions at an angle and between the obvious comfort and discomfort orientations. Additionally, in Figure 5.5.1, the frame marked as "Discomfortable 77.97%" didn't completely cover the full hand, with parts of the thumb unmarked. This could lead to incorrect identification if parts of the hand fall outside the predicted area. One approach is to provide a larger sample of training images to improve the accuracy and precision of the model's bounding box regression, which may produce tighter fits around irregular shapes like hands.

Figure 5.5.1: Thumb outside of the Hand Comfort State Identification Program's bounding box of "Discomfortable".

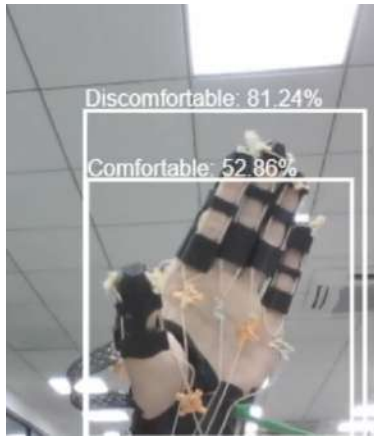

Figure 5.5.2: Ambiguous case between "Discomfortable" and "Comfortable".

f) Conclusion

Overall the hand model displayed an exemplary accuracy in correctly detecting hand gestures, providing insight for developing real-world computer vision applications. Further optimization and training surrounding borderline cases is required to refine the model for a more robust identification.

Experiment 6: Accuracy Assessment of Automatic Grab Program

a) Introduction

Even though the object detection model has an outstanding detection accuracy of 99.9%, the integration between the two systems: arduino and k210 module may introduce new variables that could impact overall system accuracy. Thus, an experiment was designed to assess the accuracy of the automatic grab program in a household environment with household objects of a water bottle, apple and remote controller. This simulates the real-world applications, allowing for a more rigorous test of the reliability of the whole system's performance. Additionally, the researcher will approach and lift the objects in a variety of angles to reinforce both the system and the experiment's reliability.

b) Material List

- Rehabilitation device
- Battery
- Phone with camera
- Plastic bottle
- Remote controller
- Apple

c) Procedure

1. Place an apple around 50 cm in front of the experimenter.
2. Wear the exoskeleton glove on the left hand appropriately.
3. Attach the battery to the motor box.
4. With the palm open, point the center of the palm to the apple. Ensure that the researcher is in a standing position and that the hand is horizontal (Figure 5.6.1).
5. Turn on the camera and start recording with the right hand. Ensure that the screen of the K210 module is visible.
6. Move your hand forwards until the palm touches the apple. The exoskeleton glove should close and wrap fingers tightly around the apple.
7. Lift the apple briefly up for 1 second.
8. Lower the apple carefully onto the table.
9. When the exoskeleton opens, move the hand away from the apple into the initial position of step 4.
10. Wait 3 seconds.
11. Repeat steps 4 to 10 four more times and change the height and grab angle.
12. Move the apple away and repeat step 1 with the bottle.
13. Repeat steps 4 to 11 two more times with the bottle.
14. Move the bottle away and repeat step 1 with the remote controller.
15. Repeat steps 4 to 11 two more times with the remote controller.

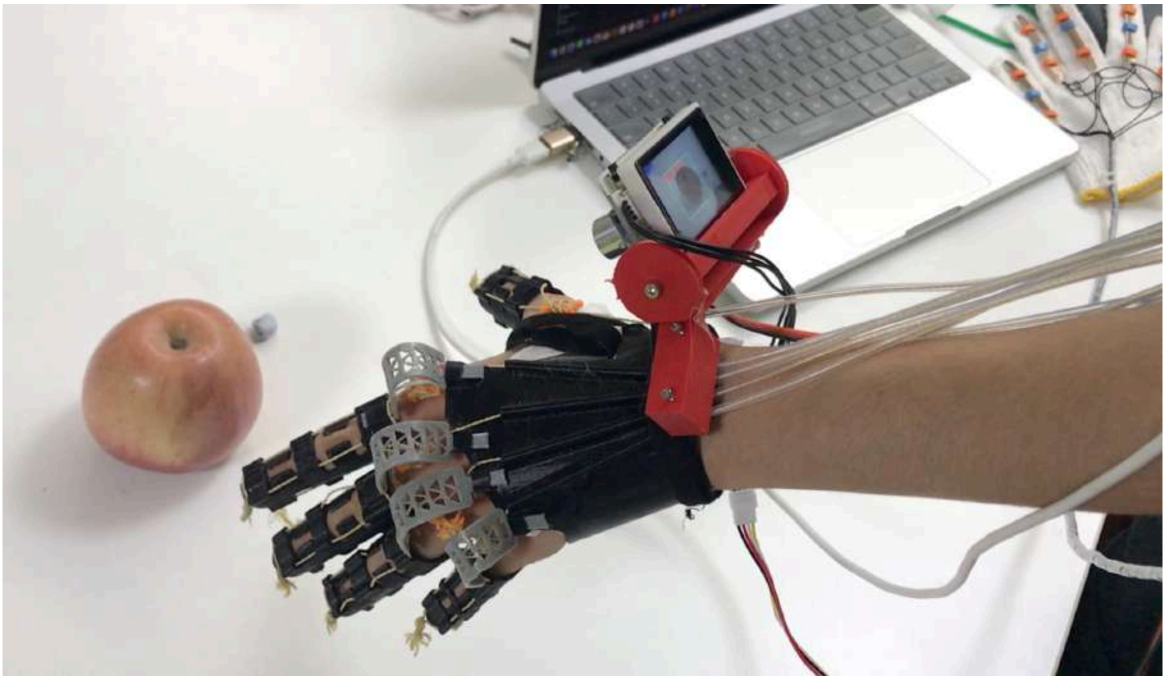

Figure 5.6.1: Horizontal hand position.

d) Data Table

Table 8: Trial versus Automatic Grab Program's object grabbing outcome supplemented with angle and height of grab

Trial	Item	Grab angle*	Grab height (relative)	Outcome	Notes
1	Apple	Top to down	High	Success	
2		Top to down	N/A	Fail	The exoskeleton activated the grab motion prematurely.
3		Horizontal	Medium	Success	
4		Horizontal	Medium	Success	
5		Top to down	High	Success	
6	Remote controller	Top to down	Medium	Success	
7		Top to down	Low	Fail	The exoskeleton

					activated at the correct time, but it was unable to pick up the remote
8		Top to down	Medium	Success	
9		Top to down	Medium	Success	
10		Top to down	Medium	Success	
11	Plastic bottle	Horizontal	Medium	Success	
12		Horizontal	Medium	Success	
13		Horizontal	Medium	Success	
14		Horizontal	Medium	Success	
15		Top to down	High	Fail	The exoskeleton glove grabbed onto the bottle's round shoulder, leading to the bottle slipping from the grasp during lifting.

*Grab angle is an evaluation of the angle formed between the original pre-grab height and the grabbing height

e) Results and Data Analysis

As shown in Table 8, a total of 15 trials across 3 objects was conducted, with 12 out of 15 trials reflecting a positive result in terms of the exoskeleton successfully detecting and picking up the object of choice, thus demonstrating an 80% reliability and accuracy. In the trials that were successful, the k210 module's bounding box regression was extremely precise such as that of Figure 5.6.1 and didn't not have any false positives for the three objects of plastic bottle, apple and remote controller. Additionally, in those successful trials, the glove was able to exert sufficient force to securely grip the object and lift it for 1 second, reinforcing the main physical composition of the glove, 3D printed flexible thermoplastic polyurethane, as both being adequately flexible for the complex flexion and extension of the human fingers as well as possessing suitable physical properties such as having a high coefficient of static friction that prevented the apple, bottle and remote controller from slipping.

However, the exoskeleton glove still possesses several failure modes shown in trials 2, 7 and 15, which can be split into 2 main categories: Control program deficiencies and mechanical design limitations. Control program failure is shown in trial 2, where the exoskeleton glove prematurely attempted to grasp the apple. After reviewing the arduino program, the main point of weakness seems to be around the sensitivity calibration of the ultrasonic sensor placed below the K210 visual module. As any slight disturbance or blockage between the ultrasonic sensor and the object may result in an incorrect reading, the trial failure may stem from this program and physical design weakness. Thus, an improved design may translate the sensor forwards to reduce unwanted disturbance. The second point of failure is mechanical design limitations. The exoskeleton failed to lift the lower section of the remote controller due to its rounded edge along with the rounded shoulder of the plastic bottle in trials 7 and 15. These points of failure with insufficient gripping around the smooth surface demonstrates the need for further optimization of the finger exoskeleton design conforming to the common geometric surfaces of everyday objects.

f) Conclusion

This preliminary testing validated the potential of this novel soft exoskeleton glove for non-invasive assistance applications. Overall, the automatic grab program demonstrated an exceptional ability to accurately grab everyday objects of an apple, plastic bottle and remote controller, achieving an overall 80% success rate. However, design changes for the ultrasonic sensor, finger exoskeleton model and algorithm refinements are required for a more mature system. Such refinements include a modification of the finger exoskeleton model's tip to increase friction during grabbing, as well as an improved system for the calibration of the camera and ultrasonic sensor to increase the object detection precision.

IV. Conclusion

The above six experiments demonstrate the proposed 3D printed glove's ability to allow patients to undergo rehabilitation effectively and in a comfortable and engaging manner. Such examples include the automatic gripping programs that achieve active assisted rehabilitation through manipulation with items in daily life and the vibration filtering algorithm that allows for the patient's recovery of the hand's fine motor functions without redundant vibrations from the motors, improving the patient's comfort. This cumulative evidence demonstrates the robust design of the rehabilitation glove and its potential to facilitate active assisted hand rehabilitation.

However, there's still a considerable potential for improvement. Such modifications include the addition of rubber paddings on the tip of the finger exoskeletons to increase friction between objects manipulated and the glove, allowing the patient to more effortlessly manipulate everyday objects and thus the glove to be more smoothly integrated into the patient's life. Additionally, the automatic grab program also displayed inconsistencies. Currently, the program consists of an arduino that utilizes a K210 module and ultrasonic sensor to detect both distance and image and the OpenCV artificial intelligence image recognition program on the K210 module trained on 700 images and 700 iterations of training to accurately detect images of everyday objects. The largest drawback to this design is the static location of the K210 module and ultrasonic sensor, overhanging from the wrist to both detect the object and the distance between the palm and the selected object. Such placement of the ultrasonic sensor may ultimately lead to some boundary cases of the sensor detecting the distance between the sensor and the index finger or thumb, leading to the premature grabbing motion resulting from the low distance detected. A possible amelioration is to move the ultrasonic sensor to an extension module added onto the finger exoskeleton and with a replacement such as one shown in figure 6.1. This modification would allow for a more accurate measurement of the distance between the desired object and the hand, improving the reliability and consistency of the automatic grab functionality of the robotic glove. However, the details of such designs are still yet to be determined in future testing and construction.

Figure 6.1: Ultrasonic sensor [6].

Lastly, the image recognition model could be further expanded to recognize a larger range of items reliably. The training image dataset of the image recognition model on the K210 module could further accumulate to approximately 2000 of a wider range of images taken from multiple angles and with a larger variety. This would increase consistency of the model's bounding box regression, thus increasing the glove's automatic grab system's overall reliability.

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Attachment:

English:

What is your age?

What upper limb rehabilitation needs do you currently have?

- Difficulty with hand/finger mobility
- Reduced grip strength
- Limited range of motion in the arm/shoulder
- Other: ________________

What rehabilitation method are you currently using?

- Traditional physical therapy sessions
- Home-based exercises
- Use of the rehabilitation device
- Combination of the above

What inconveniences have you experienced with your current rehabilitation process?

- Time-consuming/difficult to access clinics regularly
- Exercises can be monotonous and lack engagement
- Difficulty quantifying progress and adjusting exercises
- Other: ________________

Would you be willing to try a new home rehabilitation device if it could provide more effective and engaging therapy?

- Yes, I would be very interested
- Maybe, it would depend on the features and ease of use
- No, I prefer working with a physical therapist

Which of the following functions are you most concerned about in a rehabilitation device?

- Ability to independently control finger/hand movements
- Automatic detection and grasping of everyday objects
- Tracking of rehabilitation progress and adjusting exercises
- Integration with smart home technologies

How important is intelligent control (e.g. adaptive resistance, automated tasks) in a rehabilitation device for you?

- Very important - it would significantly improve the therapy
- Somewhat important - it could be helpful but not essential
- Not important - I'm more concerned about the physical design

How would you rate the comfort and wearability of the rehabilitation device?

- Very comfortable and easy to wear
- Moderately comfortable, but could use some adjustments
- Uncomfortable and difficult to wear for extended periods

Did you find the device easy to set up and use on your own at home?

- Yes, the setup and operation were straightforward
- It took some time to get used to, but I was able to use it independently
- No, I required assistance from a caregiver or therapist to use the device

How effective did you find the device in improving your hand/finger mobility and strength?

- Significantly improved my range of motion and grip strength
- Moderately improved my abilities, but more progress is needed
- Did not notice much improvement in my condition

Did you experience any issues or safety concerns when using the rehabilitation device?

- No, I did not encounter any problems or safety issues
- Yes, I experienced some minor technical difficulties or discomfort
- Yes, I had significant concerns about the device's safety and reliability

How likely are you to continue using the rehabilitation device as part of your ongoing therapy?

- Extremely likely, it has been very beneficial for me
- Somewhat likely, but I may supplement it with other methods
- Unlikely, I prefer to rely on traditional physical therapy

What additional features or improvements would you like to see in the rehabilitation device?

- More customization options for fit and personalized settings
- Integration with mobile apps or virtual reality for increased engagement
- Expanded functionality to address other upper limb rehabilitation needs
- Other: _______________

Overall, how satisfied are you with the rehabilitation device and your experience using it?

- Highly satisfied, it has exceeded my expectations
- Moderately satisfied, it has been somewhat helpful
- Dissatisfied, it did not meet my needs or expectations

Please share any other comments or suggestions you have about the rehabilitation device you have been testing:
中文

请问您今年多大了？

您目前有哪些上肢康复需求？

- 手指/手部活动受限
- 握力减弱
- 肩膀/手臂活动范围受限
- 其他：________________

您目前使用哪种康复方法？

- 传统物理疗法
- 家庭练习
- 使用康复设备
- 以上几种方法的组合

您在当前康复过程中遇到了哪些不便？

- 花费时间/难以定期前往诊所
- 练习可能单调乏味，缺乏参与度
- 难以量化进展和调整练习
- 其他：________________

如果新的家庭康复设备能够提供更有效和更有趣的疗法，您愿意尝试吗？

- 是的，我会非常感兴趣
- 也许，这将取决于其功能和易用性
- 不，我更喜欢与物理治疗师合作

在康复设备中，您对以下哪些功能最为关注？

- 能够独立控制手指/手部活动
- 自动检测和抓取日常物品
- 跟踪康复进展并调整练习
- 与智能家居技术集成

对于您来说，康复设备中的智能控制（例如自适应阻力、自动化任务）有多重要？

- 非常重要 - 它将显著改善疗法
- 有些重要 - 这可能有所帮助，但并非必不可少

- 不重要 - 我更关心外观设计

您如何评价康复设备的舒适性和穿戴性？

- 非常舒适且容易穿戴
- 适度舒适，但可能需要一些调整
- 不舒适，长时间穿戴困难

您是否发现家中自己设置和使用康复设备容易？

- 是的，设置和操作都很简单
- 花了一些时间来适应，但我能够独立使用
- 不，我需要护理人员或治疗师的帮助来使用设备

您认为康复设备在改善手指/手部活动能力和力量方面有多有效？

- 显著改善了我的活动范围和握力
- 适度改善了我的能力，但还需要更多进展
- 没有注意到我的情况有太大改善

在使用康复设备时，您是否遇到任何问题或安全顾虑？

- 没有，我没有遇到任何问题或安全问题
- 是的，我遇到了一些轻微的技术困难或不适
- 是的，我对设备的安全性和可靠性有重大顾虑

您会继续使用康复设备作为您持续疗程的一部分吗？

- 非常有可能，它对我非常有益
- 有些可能，但我可能会辅之以其他方法
- 不太可能，我更倾向于依赖传统物理疗法

您希望在康复设备中看到哪些额外功能或改进？

- 更多定制选项以适应和个性化设置
- 与移动应用程序或虚拟现实的集成，增加参与度
- 扩展功能以满足其他上肢康复需求
- 其他：________________

总体而言，您对康复设备及使用体验感到满意吗？

- 非常满意，它超出了我的期望
- 适度满意，它在某种程度上有所帮助
- 不满意，它没有满足我的需求或期望

请分享您对您正在测试的康复设备的任何其他评论或建议: